

POLYTECHNIQUE MONTRÉAL
affiliated with Université de Montréal

**Tender Response Process: Integrating Artificial Intelligence Methods Leveraging
Contractual and Competitive Data**

SALAUN ELOUAN
Department of Mathematics and Industrial Engineering

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This thesis entitled:

**Tender Response Process: Integrating Artificial Intelligence Methods Leveraging
Contractual and Competitive Data**

presented by **Elouan SALAUN**

in partial fulfillment of the requirements for the degree of *Master of Applied Science*
has been duly accepted by the examination committee composed of:

Christophe DANJOU, chair

Robert PELLERIN, member and research supervisor

Anas NEUMANN, member and research co-supervisor

Mario BOURGAULT, member

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RÉSUMÉ

La réponse aux appels d'offres constitue un mécanisme central d'obtention de contrats et un facteur déterminant de la pérennité de nombreuses organisations. Pour notre partenaire, les Commissionnaires du Québec, un acteur majeur du marché des services de gardiennage avec un revenu annuel de 747 M\$ et plus de 22 millions d'heures de sécurité travaillées en 2025, la majorité des contrats est attribuée par voie d'appels d'offres publics. La qualité du processus de réponse, et plus particulièrement la pertinence de la décision de soumissionner et de la marge proposée, conditionnent directement le taux de succès et la rentabilité de l'organisation.

Bien que ce processus génère des données potentiellement cruciales, celles-ci n'étaient jusqu'alors que peu exploitées. Les décisions reposaient principalement sur les perceptions et les expériences des gestionnaires. La revue de littérature menée dans le cadre de ce mémoire confirme que, si la modélisation tarifaire et la décision de soumission constituent des champs étudiés depuis plusieurs décennies, l'intégration conjointe de l'extraction automatisée d'information textuelle, de l'analyse de similarité entre contrats et de la modélisation tarifaire reste peu documentée. Elles demeurent inexistantes dans le secteur des services de sécurité.

Dans ce contexte, notre recherche vise à améliorer le processus de réponse aux appels d'offres en valorisant les données historiques disponibles. La démarche suit la méthodologie Design Research Methodology (DRM) et s'organise autour de cinq outils complémentaires d'aide à la décision développés en Python : (i) un module d'anticipation des appels d'offres et de fouille historique fondé sur le scraping de la plateforme SEAO et la résolution d'entité pour les contrats passés, (ii) un module de valorisation des données textuelles utilisant une architecture de génération augmentée de récupération (RAG) couplant embeddings creux, hiérarchie documentaire et BM25, (iii) un module de calcul de similarité entre contrats appris par renforcement de classement, (iv) un modèle de tarification de type Learning to Rank intégrant à la fois les caractéristiques techniques des contrats et la dynamique récente du marché, et (v) un outil de visualisation interactif assurant l'intégration de l'ensemble dans le processus métier.

Les outils ont été testés sur un échantillon représentatif d'appels d'offres publiés entre 2023 et 2026. L'algorithme de résolution d'entités atteint un taux de concordance de 92 % pour la récupération des contrats antérieurs. Le modèle de tarification, lorsqu'on se limite aux marges économiquement viables, se classe dans le top 3 des soumissionnaires dans 70 % des cas et améliore le rang du CCCQ dans 90 % des appels d'offres confrontés à la référence interne. La comparaison avec les décisions historiques sur 20 appels d'offres met en évidence un gain potentiel de profit substantiel, sous réserve d'une lecture experte des recommandations situées sous le seuil de viabilité d'environ 0,3.

Ces outils n'ont pas vocation à se substituer à l'expertise humaine, mais à l'enrichir en lui apportant une vision quantitative, structurée et systématique du contexte de chaque appel d'offres. Ils constituent ainsi un levier d'optimisation des taux de victoire et de la rentabilité, tout en facilitant la prospection et la lecture des devis techniques.

ABSTRACT

For many organizations, responding to public tenders is a central mechanism for winning contracts and a major determinant of their long-term viability. For our partner, Commissionaires Québec (CCCQ), a leading player in the security and guarding services market with annual revenues of CAD 747 M and more than 22 million security hours worked in 2025, most of the contracts are awarded through public tenders. The quality of the response process, and more specifically, the relevance of the bid/no-bid decision and the proposed margin, directly conditions the organization's win rate and profitability.

Although this process generates potentially crucial data, it has been weakly leveraged until now. Decisions relied mainly on analysts' experiences and perceptions. The literature review conducted in this thesis confirms that, while bid pricing modelling and bid/no-bid decision-making have been studied for several decades, the joint integration of automated textual information extraction, contract similarity analysis, and bid pricing modelling remains poorly documented and is particularly lacking in the security services sector.

In this context, our research aims to design a decision-support tool that leverages historical data to improve the tender response process. The work follows the Design Research Methodology (DRM) and is organized around five complementary modules developed in Python: (i) an anticipation and historical mining module based on SEO web scraping and entity matching to anticipate tenders and retrieve similar past ones; (ii) a textual data valorization module relying on a Retrieval-Augmented Generation (RAG) architecture combining sparse embeddings, document hierarchy and BM25; (iii) a contract similarity module trained through ranking-based reinforcement (RankNet); (iv) a Learning-to-Rank pricing model that jointly accounts for the technical characteristics of contracts and recent market dynamics; and (v) an interactive visualization tool ensuring the integration of all components into the business process.

The tool was evaluated on a representative sample of tenders published between 2023 and 2026. The entity-matching algorithm achieves a 92% concordance rate when retrieving previous contracts. The pricing model, when restricted to economically viable margins, ranks in the top 3 bidders in 70% of cases and improves the CCCQ's ranking in 90% of the tenders compared to the internal reference. The comparison with historical decisions on 20 tenders highlights a substantial potential profit gain, provided that an expert review is conducted of recommendations that fall below the viability threshold of approximately 0.3.

The proposed tool is not intended to replace human expertise, but rather to complement it by providing a quantitative, structured and systematic view of each tender's context. As such, it serves as a lever to optimize win rates and profitability while also facilitating prospecting and reading

technical specifications.

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LIST OF SYMBOLS AND ABBREVIATIONS

BID	Bid (offer)
CCCQ	Canadian Corps of Commissionaires – Quebec Division (Commissionaires Québec)
CoE	Centre of Excellence (<i>Centre d'excellence</i> , CDE)
LTR	Learning to Rank
RAG	Retrieval-Augmented Generation
SEAO	<i>Système électronique d'appel d'offres du gouvernement du Québec</i> (Quebec government's electronic tendering system)

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CHAPTER 1 INTRODUCTION

In many sectors of activity, winning contracts is a central issue for the growth and long-term viability of organizations. In this context, calls for tenders are a key competitive mechanism, allowing public and private organizations to select the providers that best meet their needs.

In a call for tenders, the requesting organization makes available a description and a statement of requirements for the mission it wishes to fill. The award method is also stated. Two methods predominate: award to the lowest bidder (the majority) and hybrid award, which relies on a weighting of the quality and price scores of the bid. The responding organizations then submit a priced proposal and, in some cases, a qualitative proposal. The contract is then awarded according to the designated method.

For the organization responding to tenders, each bid requires a precise costing of the human, material and financial requirements. It must ensure that carrying out the mission is feasible and provides a certain profitability. Beyond estimating the needs and costs of the mission, the success of a proposal also depends on the strategy put in place to offer a more attractive bid than the competition. This strategy must integrate multiple aspects of the market in which the organization operates while maximizing the chances of winning the contract and of profitability (Porter, 2008).

Guarding services, like many other sectors, are governed by calls for tenders. Our partner, Commissionnaires Québec (officially the Canadian Corps of Commissionnaires – Quebec Division), holds a strong position in the Quebec security services market, notably in guarding services, on which we will focus. In 2025, the organization generated revenue of \$747 M with more than 22 million hours of security work (Commissionnaires du Québec, 2026). The Commissionnaires' clients include both commercial clients and the federal government of Canada. Most contracts are obtained through public calls for tenders; the quality of the response process is therefore vital to the organization's long-term viability.

Currently, bids are built in an ad hoc manner, based mainly on the perceptions and experience of managers, even though a sufficient amount of public and internal data is available that would allow data valorization methods to be used. Digital transformation and the rise of data valorization and artificial intelligence techniques therefore represent a serious opportunity to improve decision-making and this important process. Using historical and current data on contracts and on the market makes it possible to optimize the selection and response process by providing quantified analyses that complement the experience and intuition of the experts on which current strategies rely. However, using these approaches raises several challenges. The relevant data often come from multiple

sources, have heterogeneous structures and largely consist of textual documents, such as the contracts' technical specifications. This complexity makes the data difficult to exploit with traditional analysis methods.

In this context, **the main objective of our research is to improve the tender response process by leveraging historical data.** We therefore propose a tool that provides tender management teams with the guidance needed to set a strategic price, taking into account the description of the mission as well as the external competitive environment. This tool aims to develop a more strategic, quantitative and systematic approach, increasing the chances of being awarded the contract as well as raising profitability. The model includes a contract evaluation phase based on the technical specifications and on the organization's internal history.

This thesis reports on this research project and is structured as follows. Chapter 2 presents the literature review, which surveys the main relevant documents and offers a critical perspective on them. The research methodology is then addressed in Chapter 3; building on the literature review, it sets out the specific objectives and the method used. In Chapter 4, an analysis of the existing situation presents the current tender response process. Chapter 5 presents the tool, how it works and its optimizations. The model is tested and validated in Chapter 6. Finally, Chapter 7 presents the conclusions of the project and proposes opportunities for future research.

CHAPTER 2 LITERATURE REVIEW

This chapter briefly presents the main scientific contributions on building responses to calls for tenders that leverage historical data. Before explaining our literature review protocol, we present the definitions of the key concepts used thereafter. The search protocol and the selected articles follow. Finally, we offer a critical view of the literature, highlighting the weaknesses, gaps and limitations of the contributions presented, in order to identify research opportunities.

2.1 Definition of basic concepts

This section sets out the important terms and concepts related to the subject of this thesis. A clear understanding of these concepts lays a solid foundation for better understanding the protocol presented afterwards.

2.1.1 Call for tenders

According to the *Dictionnaire de la comptabilité et de la gestion financière*, a call for tenders is defined as a “competitive procedure by which a prospective buyer of goods, supplies or services invites suppliers to submit specific proposals with a view to the award of a contract” (*Dictionnaire de la comptabilité et de la gestion financière*, version 1.2, 2006). This process places competitors in direct competition on a common basis, allowing a fair evaluation of the bids. These competitors may be any organization in the case of a public call for tenders, or selected organizations in the case of invitation-only calls for tenders. The contract is then awarded according to a method stated beforehand. In most cases, this method consists of selecting the lowest bid; sometimes, a weighting of financial and qualitative criteria may be used.

This process, mostly used by public or large entities, ensures transparency of selection and effective competition, encouraging fairer pricing.

2.1.2 Bidders

A bidder is an organization competing for a contract that submits a bid in a call for tenders. Bidders can be of different types and sizes and must make sure they are capable of honouring the contract. Consequently, they may choose to withdraw if they consider the contract infeasible or unprofitable. Each bidder wishing to bid must analyze the requirements of the mission to be carried out and

prepare a financial estimate. Then, depending on the strategy it wishes to adopt to ensure the success of its bid, it builds its offer by choosing, in particular, the margin it will apply.

2.1.3 Response strategy

The response strategy is the key element that influences both the success of the bids submitted and their profitability. A first step of this strategy consists of deciding whether or not to bid on a call for tenders. This decision must of course consider response capacity in terms of resources, but also the competitive environment of the call for tenders. Indeed, depending on the number of competitors, their nature, the history of the contract or the internal context of the organization, it may be wiser to let the contract go and avoid, for example, a needless mobilization of resources. Then, once the decision to bid has been made, the strategy must make it possible to set the price to submit, and in particular the margin to apply on top of fixed costs. This price must make it possible to win the contract while generating an attractive profit. More specifically, in a lowest-bidder award, one tries to win the contract at the highest possible price, i.e., close to the second-lowest bid, the difference being a loss for the winner: the “winner’s curse”. This strategy must therefore be developed taking full account of the competitive environment.

2.2 Search protocol

Because of the multidisciplinary nature of our field of study, we use the Scopus scientific database. This database was chosen for its great diversity, offering more than 100 million scientific contributions.

To build our search protocol, we must identify the main concepts related to our study. Since the field of calls for tenders for security contracts alone is too narrow to yield a sufficient number of scientific contributions, we broaden our search to wider sectors that use tendering procedures (notably construction, engineering and public procurement). We therefore look for data-driven methods for studying contracts in the context of calls for tenders. The concept map listing the concepts and their synonyms is presented in Table 2.1.

Table 2.1 Key concepts of the literature review

Call for tenders	Response strategy	Data-driven
Tender Procurement Bidding Bids	Decision Evaluation Strategy	Data-driven Historical data Machine learning Artificial intelligence Natural language process

Using these concepts, we build a first query: TITLE-ABS-KEY ((tender* OR procurement OR bidding OR bids) AND ("data-driven" OR "historical data" OR "machine learning" OR "artificial intelligence" OR "NLP" OR "natural language processing") AND (decision OR evaluat* OR strateg*)).

After analyzing the first results, we decided to add keywords and exclusions, notably on real-time auction markets for energy or online advertising, in order to refine our set of results. Our final query is then: TITLE-ABS-KEY ((tender* OR procurement OR bidding OR bids) AND ("data-driven" OR "historical data" OR "machine learning" OR "artificial intelligence" OR "NLP" OR "natural language processing") AND (decision OR evaluat* OR strateg*) AND (contract) AND NOT (energy OR electric* OR advert* OR "ads"))

The query, run on March 17, 2026, returns 187 results. As illustrated in Figure 2.1, we obtain a sample of 169 documents by applying filters on document type and language to exclude documents that are not articles or conference papers in English or French. A first reading of the article titles alone then eliminated 116 documents, mostly contributions on related but irrelevant topics for our study, such as corruption detection, auctions on financial markets or the management of calls for tenders from the buyer's point of view. Finally, reading the abstracts and, where applicable, the full articles eliminated 28 further publications. In the end, 24 relevant publications, presented in Table 2.2, were retained. They are described in the next section.



Figure 2.1 Diagram of the literature review

2.3 Results

The articles identified address distinct sub-parts of our subject. We can identify a few common themes: Tendering 4.0, the modelling of decisions and strategies, contractual information extraction, and contractual risk analysis using artificial intelligence.

2.3.1 Tendering 4.0

The emergence of new 4.0 technologies has transformed the tendering process, giving rise to a digitalized and reorganized process with new needs and new opportunities. Simões et al. (2023) propose a systematic literature review on what they call “Procurement 4.0”, identifying key technologies such as artificial intelligence and ERP systems. This transformation is particularly visible in the optimization of tendering processes, where the integration of artificial intelligence (AI) and software robotics now makes it possible to automate complex tasks such as document generation or the preliminary evaluation of bids, notably thanks to natural language processing (Cusumano, Rempling, et al., 2022). AI is becoming a major decision-making lever; indeed, machine learning and natural language processing (NLP) algorithms show remarkable capabilities, drastically reducing analysis time while surpassing the accuracy of human evaluation (Park et al., 2021). These techniques can be useful in the qualitative evaluation of the risks associated with contracts, using NLP techniques to, for example, produce a reliable risk assessment. Malatesta and Mancini (2026) set out a framework and identify the corresponding approaches. Finally, these techniques can serve as bidding-strategy support tools: applying predictive models coupled with artificial intelligence makes it possible to isolate the determining variables of the tender response process and to provide visibility on the optimal price ranges for winning the contract (Molina-Miranda et al., 2023).

Table 2.2 Selected articles

Authors	Year	Title
Biruk et al.	2017	Modelling contractor’s bidding decision
Charbel et al.	2025	Predicting the Level of Competition and Determining Optimal Bidding Strategies for Bundled Projects
Chisala	2017	Quantitative Bid or No-Bid Decision-Support Model for Contractors
Choi & Lee	2022	Contractor’s Risk Analysis of EPC Contracts Using Ontological Semantic Model and Bi-LSTM
Cusumano, Saraiva, et al.	2022	Intelligent building contract tendering – potential and exploration
Cusumano, Rempling, et al.	2022	Natural language processing as work support in project tendering
Dawood	1994	Developing an integrated bidding management expert system for the pre-cast concrete industry
Faraji et al.	2022	A Bid/Mark-Up Decision Support Model Based on Project Complexity in the Petroleum Industry

Authors	Year	Title
Kim et al.	2020	Application of Natural Language Processing (NLP) and Text-Mining of Big-Data to Engineering-Procurement-Construction (EPC) Bid and Contract Documents
Król et al.	2024	Application of ML Algorithms for Identification of Key Criteria Groups in Public Tendering
Lops et al.	2022	AI-based Decision Support Systems for the Management of E-procurement Procedures
Malatesta & Mancini	2026	AI-Enabled Contractual Knowledge and Risk Governance in EPC Projects
Mencía et al.	2013	Using Data Mining on Linked Open Data for Analyzing E-Procurement Information
Molina-Miranda et al.	2023	Application of explainable AI to analyze basic features of a tender
Moriyani et al.	2026	Beyond One-Size-Fits-All: A Novel Data-Driven Framework for Quantifying Bidding Competition Intensity
Padhy et al.	2021	Application of NLP to Automatically Identify Exculpatory Clauses in Construction Contracts
Park et al.	2021	A Digitalized Design Risk Analysis Tool with ML Algorithm for EPC Contractor's Technical Bid
Siciliani et al.	2023	AI-based decision support system for public procurement
Simões et al.	2023	Unlocking the Potential of Procurement 4.0: Digitalization, Industry 4.0, and Information Systems
Sturm et al.	2026	Augmenting Firm Diversification Behavior Prediction with Graph Embeddings
Tirmizi & Arif	2022	Conceptual Approach for the Use of AI for Contractual Risk Assessment in Infrastructure Projects
Vásquez & Kumral	2025	ANN and SVM for more accurate cost estimation in underground mining
Young-Hun & Gitae	2024	Analysis Bidding Price Analysis Using K-Means Cluster, ML, and Deep Learning
Zhang et al.	2026	Automated multi-label classification of risk clauses in construction contracts using GPT-driven data

2.3.2 Modelling bidders' decisions and bidding strategies

This theme brings together contributions on the modelling of bidding decisions, including the Bid/No-Bid decision, price determination and the analysis of the competitive environment.

2.3.2.1 The “Bid or No-Bid” decision process

The decision whether or not to bid on a call for tenders (“Bid/No-Bid”) is the first trade-off in a response process. It is important, since it involves selecting the attractive tenders while avoiding wasting resources on others that are less attractive or futile. Historically, most contractors have tended to approach this decision purely intuitively (Chisala, 2017). The literature, however, stresses the need to rationalize this process.

As early as 1994, Dawood (1994) reviewed the needs and the promising methods. He proposed an

expert system, based on intelligent rules derived from previous contracts, in order to move away from intuition-based methods. More recently, methods have developed and the decision is now considered a true multi-criteria problem. In their publication, Biruk et al. (2017) propose using AHP (Analytic Hierarchy Process) for multi-criteria weighting, combined with linear programming, to determine the bid price and the degree of desirability of the contract. Along the same lines, Chisala (2017) models this attractiveness assessment by weighting 75 internal and external factors to compute a score that can be adapted to the company's risk tolerance and objectives. Going further, finer analyses of the project's intrinsic characteristics can be considered: Faraji et al. (2022) introduce a project complexity index as a key determinant of the bidding decision.

2.3.2.2 Competitive bidding model

The applied margin (or mark-up) is the final and decisive component of the bidding strategy. This margin is added to direct costs and possible provisions; its level, which is strategic, must take into account the company's objectives as well as the competitive environment (Vásquez & Kumral, 2025).

The integration of qualitative variables into the margin calculation is a recent topic. Older models such as those of Gates or Friedman (1956; 1967), which are strictly mathematical, have been improved by expert systems based on managerial experience (heuristic rules) to weight factors such as the client's reputation or the economic context (Dawood, 1994).

Recent approaches differ according to whether they focus on the intrinsic attributes of the contract or on the dynamics of the market in which the bid takes place. At the level of the individual contract, several works propose calibrating the margin according to the project's own characteristics. Biruk et al. (2017) propose adjusting the margin by interpolation according to an overall attractiveness score, so that it varies between two justifiable bounds. Faraji et al. (2022) show that margins, whether financial or time-related, can be correlated with a project complexity index using non-linear regression models, making it possible to identify viable margin thresholds. In mining, García Vásquez and Kumral (2025) compare artificial neural networks (ANN) and support vector machines (SVM) for estimating the costs of underground projects from the contractor's point of view. Beyond accuracy, processing is 80 times faster than manual estimation, which is a considerable advantage during the bid preparation phase. At the market level, other works focus on the competitive structure and on the aggregate behaviour of bidders to guide pricing. Young-Hun and Gitae (2024) apply K-Means clustering, followed by machine learning models (decision tree, random forest, SVM, KNN) and deep learning (multilayer perceptron), to the analysis of bid prices in public procurement, showing that expected prices fall within an 88% range, a sign of stable

bidding behaviour within each group of projects. Charbel et al. (2025) integrate machine learning algorithms (Random Forest, ANN, KNN, Decision Tree, SVM, GBM, LightGBM, CatBoost) into Friedman’s and Gates’ probabilistic models to predict the level of competition and determine optimal bidding strategies for bundled projects. By selecting variables with the Boruta algorithm and using Monte Carlo simulations, this framework identifies the key variables influencing competition (project duration, estimated amount, month of publication) and allows contractors to quantify win probabilities and expected profit.

2.3.2.3 Competitiveness factors

Beyond contractual attributes, competitive dynamics are also linked to the very structure of the market. Assessing the competitive environment is a significant issue in developing the strategy.

In this respect, Charbel et al. (2025) show that bundling projects restricts competition to large companies with strong resource capacities. Moreover, the authors show how market saturation forces companies to adopt aggressive pricing strategies, significantly reducing margins. Moriyani et al. (2026) propose an approach based on graph neural networks (GNN) to model the relationships between competitors. The intensity of competition is then quantified and shown to be intrinsically linked to the specific attributes of the contract, providing data-driven competition metrics that go beyond a simple count of the number of bidders. Also using graph models, Sturm et al. (2026) use graph embeddings via the Node2Vec algorithm to predict the diversification behaviour of companies in public markets. This approach provides insights into preferred contract types and into the strong sectors of the various market players. Taking a more global view of the market, Siciliani et al. (2023) and Lops et al. (2022) develop anomaly detection systems by identifying critical risk indicators such as a hyper-concentrated market structure, the absence of historical players or the presence of superfluous bidders submitting artificial bids. Finally, Król et al. (2024) address the standardization of evaluation criteria in public procurement. By applying natural language processing and a clustering algorithm, they analyze and compare evaluation criteria across different procurement procedures, thereby giving bidders a clearer understanding of the evaluation criteria.

2.3.3 NLP for contractual information extraction

Data collection and extraction are crucial for feeding the methods described above. Structured market data may be available directly on open data platforms (Moriyani et al. (2026); Faraji et al. (2022)) or through web exploration techniques using “crawlers” (Król et al. (2024); Lops et al. (2022)). However, much of the information about the contracts themselves is only available in unstructured form in PDF documents. A pre-processing phase is essential: Choi & Lee (2022)

and Park et al. (2021) stress the structure of these files, which requires removing noise and using parsers coupled with optical character recognition (OCR). Kim et al. (2020) use domain-specific semantic lexicons to extract clauses and classify them. Padhy et al. (2021) rely on tokenization and rule-based tools (PhraseMatcher) to extract key information.

Advanced extraction is moving towards semantic and natural language processing (NLP) approaches. Mencía et al. (2013) develop recursive algorithms to convert semantic data into relational formats. Vectorization models such as Sentence-BERT are used by Król et al. (2024), and Zhang et al. (2026) then explore the potential of large language models (LLMs). This automation is necessary for building the standardized databases required to develop high-performing models. High-quality standardized historical data are a significant but essential challenge for making optimal use of data (Cusumano, Saraiva, et al., 2022).

2.3.4 Contractual risk analysis using artificial intelligence

The automatic identification and assessment of contractual risks is a major research area aimed at improving the quality of bidding decisions.

Kim et al. (2020) made the first attempts to classify extracted clauses using semantic rules. Next, Park et al. (2021) developed a design risk analysis tool for evaluating technical specifications during bidding in an EPC (Engineering, Procurement & Construction) context. Their system includes two machine-learning-based modules: technical risk extraction (TRE) and design parameter extraction (DPE). The TRE module achieves an accuracy roughly 50% higher than manual extraction, while the DPE module considerably reduces analysis time compared with manual review by an engineer, with risks classified into five severity levels. Also in the context of EPC contracts, Choi & Lee (2022) combine an ontological semantic model with bidirectional long short-term memory networks (Bi-LSTM) for risk analysis. More recently, Zhang et al. (2026) used the generative language models GPT-2 and GPT-4 for data augmentation in the multi-label classification of risk clauses in construction contracts.

More broadly, Tirmizi and Arif (2022) propose a conceptual framework based on artificial neural networks (ANN) for assessing contractual risks in petroleum infrastructure projects. Their approach relies on three databases (risks, impacts, mitigation strategies) and identifies contractual risks. The system provides a ranking of risk impacts and corrective solutions, addressing organizational problems related to working in silos and a lack of expertise in contract drafting.

2.4 Critical analysis

The analysis of the literature reveals a significant evolution in the methods used to support the tendering process. The first works, dating from the 1990s, rely on expert systems and classical multi-criteria methods. Then, the 2010–2020 period saw the emergence of data mining methods and quantitative decision models. Finally, from 2020 onwards, a marked acceleration can be observed with the adoption of NLP, deep learning and language models, as well as graph architectures.

Three main categories of approaches can be distinguished. Classical supervised learning methods (random forest, SVM and neural networks) are widely used for price prediction, classification and cost estimation, often with remarkable performance. Next, NLP and text processing techniques are becoming increasingly sophisticated, moving from approaches based on semantic rules to transformer models such as GPT. Finally, graph-based approaches are emerging as an innovative category, making it possible to model the structural relationships between market players and contracts.

Despite the significant advances in the literature, several limitations and gaps deserve to be highlighted. First, the fragmentation of approaches. Most contributions focus on a single aspect of the tendering process in isolation: either the Bid/No-Bid decision, cost estimation, or document analysis. Very few works propose an integrated framework covering the entire response cycle, from detecting the opportunity to formulating the price. Only Lops et al. (2022) and Siciliani et al. (2023) adopt a systemic approach, but their work remains centred on the context of Italian public procurement. This fragmentation is compounded by a generalizability problem: many contributions are anchored in a specific sector (construction, EPC, mining) or in a particular geographical context (Korea, Poland, Ecuador, Italy), which makes the transferability of the models uncertain. The services sector, and a fortiori the security sector, remains largely under-represented, even though it has notable specificities, such as the strong predominance of labour costs, the recurrence of contracts and a very low proportion of variable costs (notably compared with the construction sector).

Next, data processing. On the one hand, integrating multimodal data remains a challenge; NLP methods excel at processing textual documents, while predictive models exploit structured data. However, the actual response process involves merging these two types of sources, to which competitive information and the teams' tacit knowledge must be added. On the other hand, all the works surveyed deal with English-language corpora, whereas our operational context relies on contracts written in French, which poses additional challenges in terms of available language resources and pre-trained models. Furthermore, the temporal and dynamic dimension of the market is often neglected; existing models generally assume a static environment, whereas market conditions, competitors' strategies and clients' needs change over time. Only Sturm et al. (2026) propose

a temporal validation using expanding windows, but this approach remains limited.

Finally, practical considerations related to deployment are rarely addressed. Most contributions remain at the prototype or proof-of-concept stage, without evaluating their integration into existing operational processes. Cusumano et al. (2022) are an exception with their evaluation with end users, which reveals concerns about the tool's reliability and data quality.

A summary of this critical analysis is presented in Table 2.3.

Table 2.3 Summary of the critical analysis

Authors	Characteristics of the methods					
	Bid decision	Price/margin decision	Technical data	Market data	Data extraction	Temporal variables
(Biruk et al., 2017)	✓	✓	✓			
(Charbel et al., 2025)		✓	✓	✓		
(Chisala, 2017)	✓		✓	✓		
(Choi & Lee, 2022)			✓		✓	
(Cusumano, Saraiva, et al., 2022)					✓	
(Cusumano, Rempling, et al., 2022)					✓	
(Dawood, 1994)	✓	✓	✓	✓		
(Faraji et al., 2022)	✓	✓	✓			
(Vásquez & Kumral, 2025)		✓	✓			
(Kim et al., 2020)			✓		✓	
(Król et al., 2024)					✓	
(Lops et al., 2022)			✓		✓	
(Malatesta & Mancini, 2026)					✓	
(Mencía et al., 2013)				✓	✓	
(Molina-Miranda et al., 2023)		✓		✓		
(Moriyani et al., 2026)	✓			✓		✓
(Padhy et al., 2021)			✓		✓	
(Park et al., 2021)	✓		✓		✓	
(Siciliani et al., 2023)				✓	✓	
(Simões et al., 2023)						
(Sturm et al., 2026)				✓		✓
(Tirmizi & Arif, 2022)			✓			
(Young-Hun & Gitae, 2024)		✓	✓	✓		
(Zhang et al., 2026)			✓		✓	

2.5 Conclusion

This literature review has provided a comprehensive picture of the scientific contributions on the application of artificial intelligence and data-driven methods to the tendering process. The analysis of 24 articles published between 1994 and 2026 reveals three main research axes: the digitalization of procurement processes and the development of integrated decision-support systems; the formal modelling of bidding decisions and pricing strategies; and the automatic processing of unstructured data from contractual documents.

Nevertheless, our critical analysis reveals significant gaps: fragmentation of approaches, lack of integration between the different steps of the process, limited generalizability, absence of the services sector, and scarcity of empirical validation in an operational context. These findings motivate and guide our research, which aims to develop an integrated decision-support framework jointly exploiting automated information extraction, historical data analysis and the formulation of pricing strategies, all validated in the real context of the security contracts of Commissionaires Québec. The next chapter describes the method used in this research project.

CHAPTER 3 RESEARCH METHODOLOGY

3.1 Introduction

This chapter describes the research approach used. We first set out our specific objectives, then present the methodological approach adopted to achieve them.

3.2 Research objectives

The general objective of our research is to improve the tender response process by seeking to leverage historical data, notably by integrating artificial intelligence and data valorization methods. To achieve this, several specific objectives were defined:

- Characterize the decision variables and indicators that structure bidding decisions;
- Determine which models and approaches from the literature can be transposed to the partner's specific context; and
- Adapt the selected approaches and validate their viability in a real context by applying them to our partner's historical data.

3.3 Research methodology

This research project aims to develop tools to facilitate and improve the management of tender responses for guarding services contracts. The tool and its associated models will support managers at different points of the response process, from the publication or pre-publication of a call for tenders to establishing the financial scenario to favour in the bid. This tool must make it possible to take into account multiple decision indicators, about both the contract and the competitive environment. To do so, we must develop the capacity to process and integrate raw, multimodal data, as well as models that ultimately extract viable and relevant information.

The methodology adopted to conduct this research is the Design Research Methodology (DRM), developed by Blessing and Chakrabarti (2009). This choice is justified by the nature of the intended contribution: a tool whose usefulness must be demonstrated both scientifically and operationally. This method allows us to assess the benefits of our contribution without requiring full implementation. DRM structures research into four sequential and iterative phases, illustrated in Figure 3.1. First, the literature review makes it possible to clearly define the research objectives and delimit the

functional scope of the tool. Next, the descriptive analysis of the case study aims to understand the current state of the phenomenon studied and the available data. Then, the prescriptive study is the core of the design and development work on the tool and its elements. Finally, a second descriptive analysis makes it possible to evaluate the tool developed by using it in different situations of our case study.

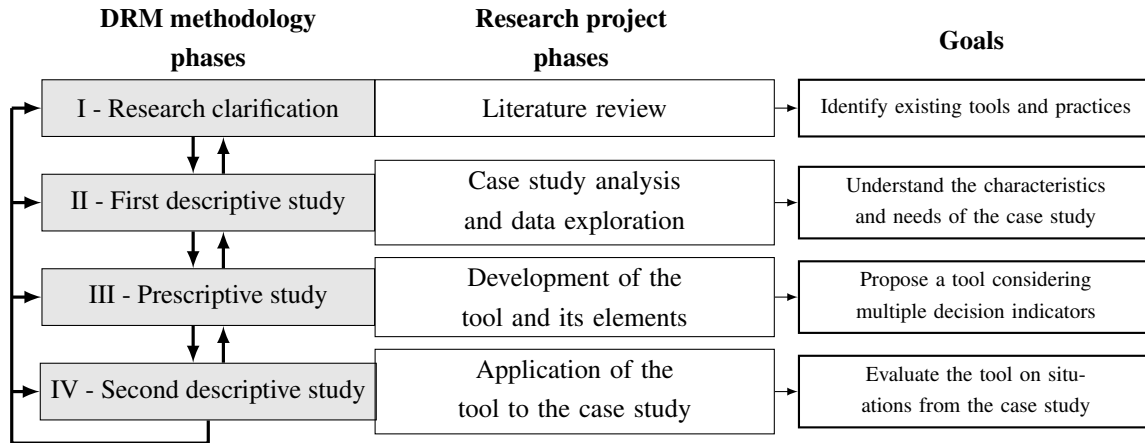


Figure 3.1 Structure of the DRM methodology, adapted from Blessing and Chakrabarti

In the previous chapters, we carried out the research clarification phase by means of a systematic review, allowing the exhaustive identification of current practices related to our subject. The critical analysis made it possible to expose the limitations of these practices and to perceive the gaps that constitute research opportunities.

Chapter 4 presents the case study analysis. The process is modelled using the BPMN 2.0 formalism, suited to modelling data interactions and decision-making. This modelling makes it possible to identify the weaknesses of the process and where the potential contributions of our research can be integrated. The external data used come from the open database and the website of the Quebec government's electronic tendering system (SEAO); filtering was applied to keep only calls for tenders related to guarding services, in their latest update. The internal data used come directly from the contractual documents posted online by clients when calls for tenders are published and downloaded by the partner; the extracted data constitute the technical characteristics of the contracts.

The tools developed are presented in Chapter 5, where the iterative nature of the DRM methodology allowed us to continuously assess the elements needed for our final tool. Using entity resolution methods allows us to retrieve past contracts despite variations in the identification fields. Developing a retrieval-augmented generation (RAG) tool makes it possible to grasp the subtleties of a contract in order to systematically extract and compute contract variables directly from the contractual document. To retrieve technically similar contracts, a simple neural network was initialized

and then retrained from human rankings in order to capture the characteristics that really matter according to managers' experience. Next, a Learning to Rank pricing model makes it possible to rank potential margins according to their optimality and chance of winning the call for tenders; this XGBoost-based model is consistent with the relatively small size of our database. Finally, a graphical interface is developed with the Python library Streamlit, allowing easy deployment in the partner's environment and accessible use by managers.

We address the application and results of our tool in Chapter 6. To evaluate our tool and the different elements developed during our research, we can rely on our partner's historical data for the last five years, as well as on open data published by the government. We update our data throughout the project in order to build a dataset as close as possible to current events. The entity resolution algorithm is evaluated on the proportion of true positives returned for contracts from 2025. We then apply our tool to situations, i.e., publications of more or less recent calls for tenders, in order to compare the validity of the models implemented, but also their performance relative to our partner's actual behaviour. Performance is studied by taking into account the rank, i.e., the place at which the bidder finishes, and the profit generated. We use 57 contracts published in 2023 and 2024 to test the general behaviour of our model. Finally, we study 20 contracts published between 2023 and today, contracts in which our partner participated, in order to validate and compare performance between our model and the partner's history. The data used in these experiments are data that predate the study by at least a certain time, in order to account for the time it takes for the results of a call for tenders to be published.

3.4 Conclusion

The methodology applied during our research project is the DRM methodology. This method allows progressive validation of our tools without requiring full operational deployment. We propose a framework of tools integrating multiple decision indicators. This framework is a complete environment ranging from the collection and transformation of raw, multimodal data to the computation and visualization of the indicators relevant to the manager during the tender response process. The next chapter presents the case study and its specificities.

CHAPTER 4 ANALYSIS OF THE EXISTING SITUATION

In this chapter, we present our partner, the case study, the existing process and the data on which we will rely to develop our tool. We begin with a general presentation of our partner, then of the case study, in order to grasp the constraints and context of our project. Next, we describe the current tender response process in order to identify possible improvements. Finally, we present the usable data and their specificities.

4.1 Presentation of the partner

This section presents the partner associated with this project, its field of activity and its organization, in order to clarify the context of our research.

Our partner, Commissionnaires Québec (officially the Canadian Corps of Commissionnaires – Quebec Division), is a century-old non-profit security organization whose social mission is to support veterans. It is a leading player in security services in Quebec, notably in guarding and patrol services for both public and private entities, with more than 22 million hours worked and \$747 M in revenue in 2025 (À propos de Commissionnaires, 2026).

The Commissionnaires' office is organized into several divisions. The various entities of these divisions may intervene at different levels of the process.

The Centre of Excellence (CoE), a component of the finance division, is responsible for steering the process of renewing or acquiring contracts arising from calls for tenders. Made up of several units, this sub-division manages the process from prospecting opportunities to building the response forms containing the defined financial offer and the various requirements (legal documents, references, proof of technical capacity, etc.).

For the two crucial decisions of the process, namely the decision to bid and the choice of the margin to apply, a multi-division panel is consulted. This panel is made up of directors from operations, finance, human resources and other divisions or sub-divisions if necessary.

4.2 Case study

In this section we present the specific case study of our research: the tender response process, a crucial process for our partner's long-term viability.

4.2.1 Calls for tenders

We are interested in the tendering process. It is particularly important for the Commissionaires, since the overwhelming majority of guarding contracts in Quebec are awarded through this process. The various clients, whether public (municipalities, ministries, school service centres, etc.) or private (stores, offices, museums, etc.), publish on the government platform *Système électronique d'appel d'offres du gouvernement du Québec* (SEAO) the contractual documents (rules, contract, specifications, bid form) related to the contract they wish to fill. Depending on the type of call for tenders, open or by invitation, different bidders are free to submit a bid if they meet the criteria. A period of about one month allows them to prepare it. During this period, bidders may ask questions about the contract. If clarifications or changes are made, addenda are published. Criteria such as holding certain certifications, having carried out similar missions for other clients or proving the technical capacity to carry out the mission may be required to participate. Then, after bidders submit their bids, the bid opening results are published. Finally, the contract is awarded according to terms specified beforehand: in rare cases with a weighting of financial and qualitative criteria, but more generally to the lowest financial bid. Sadoune et al. (2023) also use these same SEAO data to train lowest-price auction simulation models. This process is described in Figure 4.1.

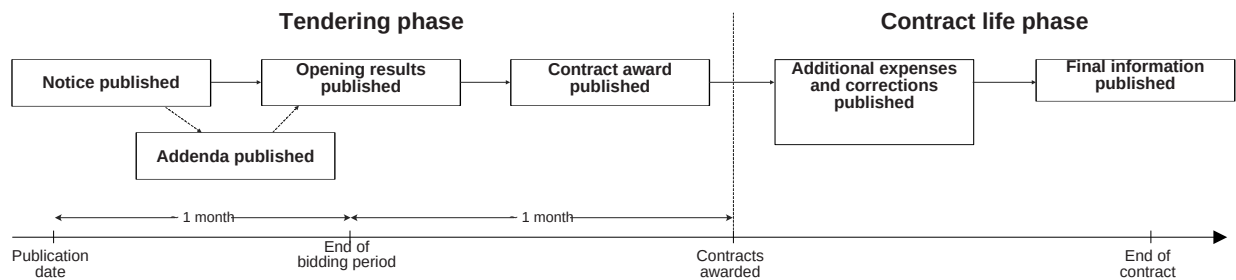


Figure 4.1 Publication process of a call for tenders on SEAO

4.2.2 Contracts

The contracts we study are guarding and patrol contracts. In these, bidders must provide officers to ensure the security of a site (stores, offices, school service centres, health facilities, etc.). Sometimes there may be multiple sites and on-call requests; this is the case for patrols, where officers must travel to the site of a call or an alarm.

The technical specifications to be considered for a contract are defined in the technical specifications document provided by the client. They include:

- The schedules and volume of hours worked by the officers;

- The bonuses to be paid to the officers;
- Material requirements: uniforms, equipment and vehicles;
- Mandatory training hours; and
- Possible penalties.

The cost of labour, i.e., the officers' hourly rate, represents the largest share of the contract price. It is often this amount that the client asks bidders to enter in the price schedule when responding to a call for tenders. In Quebec, the *Decree respecting security guards*, D-2, r. 1 (Act respecting collective agreement decrees, 2026) governs the minimum hourly rate and hourly bonuses of officers; a legal floor price can therefore be computed for each contract. It is this price that is used to define the margin ratio (Eq 4.1). The breakdown of costs in the total hourly rate per officer is shown in Figure 4.2.

$$\text{Margin ratio} = \frac{\text{Total price submitted}}{\text{Minimum price regulated by the decree}} \quad (4.1)$$

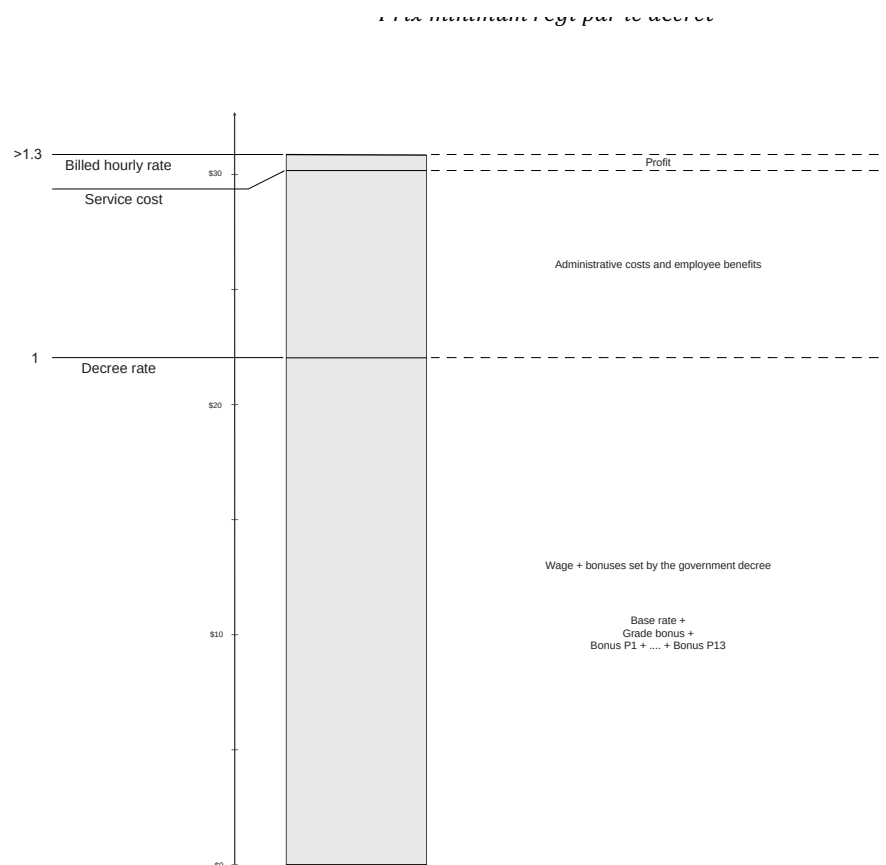


Figure 4.2 Breakdown of costs in an officer's hourly rate (indicative scale)

In general, the contracts are very similar, particularly with regard to the mission entrusted and the method used to determine the price to propose.

4.3 Process in place at the Commissionaires

The processes currently in place for responding to calls for tenders at the Commissionaires are presented in Appendices A, B, C and D using the BPMN 2.0 formalism (“Business Process Model and Notation (BPMN), Specification Version 2.0”, 2011), the most suitable reference formalism in our case, allowing data and decision-making to be taken into account (Mosser et al., 2022).

Appendix A describes the overall process, made up of the main steps from the identification of a potential contract to the conclusion of the response process. Four main steps can be distinguished. The pre-analysis leads to the first major decision of the process: whether or not to bid on the call for tenders under consideration. If the decision to bid is made, the following steps are linked: preparation of the analysis, study of financial scenarios and finalization of the bid file.

The pre-analysis, described in Appendix B, must allow decision-makers to determine whether the call for tenders under consideration is viable and feasible given the available resources. The Centre of Excellence (CoE) carries out this pre-analysis; the first characteristic information about the contract (number of hours, duration, client, etc.) is documented, as well as the constraints and elements that could have financial impacts. During this process, the CoE may exchange with the client to resolve potential questions; these are generally clarified through addenda published by the client on the SEAO online platform. The summary of the information documented during this pre-analysis is then presented to an executive committee made up of a panel of directors from operations, finance, human resources and other divisions or sub-divisions if necessary. The decision whether to proceed is then made.

If the decision to pursue the call for tenders is made, the process of preparing the response begins. During the analysis (Appendix C), all the elements and costs related to the contract (hours, bonuses, resources, insurance, training, etc.) are listed and then calculated. At this point, we have the gross cost of the contract; it then remains to determine the margin to apply to obtain the price to submit to the client. This is done during the financial scenario study step (Appendix D). The aim is to find the margin to apply that meets market expectations and ensures the contract’s profit. To do this, the CoE director can rely on some data, as well as on his or her experience of the market. When the contract is substantial, the executive committee is consulted to establish specific guidelines to apply. Once the margin is chosen, our financial offer is finalized and the price schedule to submit to the client can be filled in. A last step of finalizing the file takes place, during which the qualitative offer, made up of documents meeting the client’s requirements, is built if needed. The response process ends with the submission of the bid to the client.

The process described above has some shortcomings:

- Weak integration of decisions based on precise and systematic indicators that take the context into account, whether that of the contract or of the market. Mostly, it is the managers' intuition and experience that prevail in decisions;
- Access to the available data is difficult. They are stored in a cloud environment in a disorganized way, which makes them hard to find, even when they exist. Only the data of contracts studied in depth by the Commissionaires are kept. The manager must therefore remember whether a piece of data might exist in the database, then search for it manually and inefficiently, which discourages its use;
- The lack of systematic and unbiased tracking of results hinders the implementation of mitigation strategies. Lessons learned are poorly managed and difficult to take into account because transposing results to other contracts seems unclear; and
- There is no defined and automated process for retrieving new opportunities. Prospecting is done manually and without anticipation, whereas anticipation could allow a better preparation or lobbying phase, strengthening the partner's position to win the contract.

4.4 Available data

To develop our tools, we first carried out an in-depth analysis of the various databases on which we could rely.

4.4.1 External data

A first important database is the platform of the Quebec government's electronic tendering system (SEAO). This platform hosts the vast majority of the tendering processes in which the Commissionaires participate. The platform publishes weekly and publicly, via Open Canada, the results of past and ongoing calls for tenders.

Although the data published in semi-structured form (JSON) can already be of considerable use, we instead rely on these publications to retrieve the direct URLs of the web pages for each contract (which are not directly accessible from the site). These pages list much more information, which can be extracted using web scraping techniques. This technique can also be used in the same way to automatically fetch new publications or anticipate potential opportunities. The diagram of this collection process is shown in Figure 4.3.

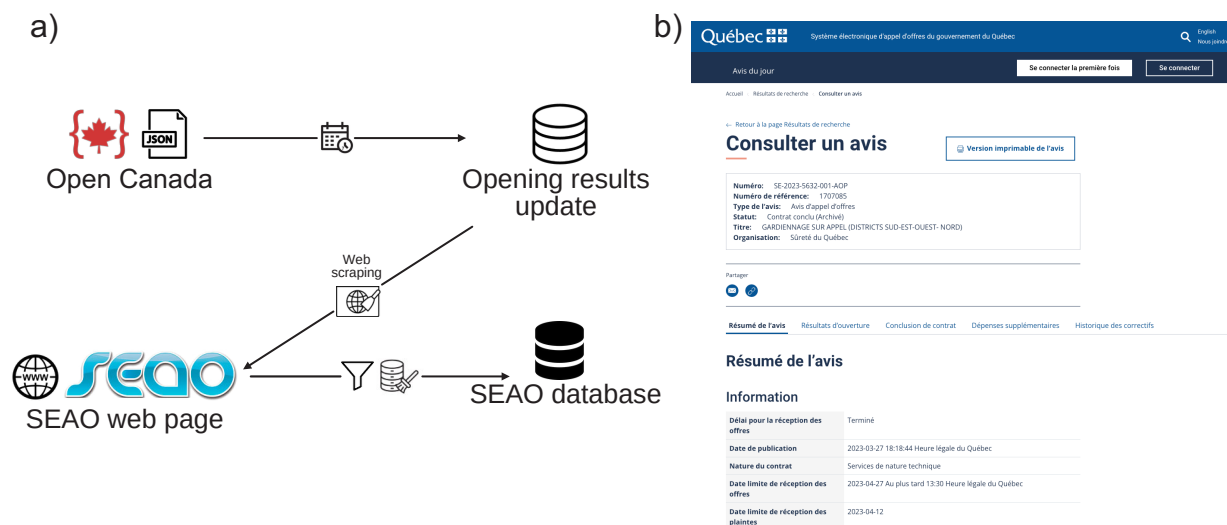


Figure 4.3 a) Diagram of SEO data retrieval; b) Example of a call-for-tenders web page on SEO

The Open Canada database publishes all Quebec calls for tenders; we therefore apply filters on titles and on UNSPSC codes (*United Nations Standard Products and Services Code (UNSPSC)*, 2026). Since each new publication in the database may only represent an update of the status of a call for tenders, we ultimately keep only the most recent status.

The data retrieved from the SEO web pages include: Notice summary; Bid opening results; Contract award; Additional expenses; Final information; and Correction history (sub-menus of the page). The main useful information extracted is listed non-exhaustively in Table 4.1.

We thus build a database of 1,960 publications published between 2015-01-01 and 2026-03-27. Of these, 94% (1,836) of contract awards are available, and for open or invitation-only calls for tenders, which represent 49% (957), 95% (908) of the results are available.

Table 4.1 Data extracted from the SEO platform

Menu	Field	Description
General information	Number	Unique client number of the contract
	SEO reference number	Number in SEO; may be the same for some contract renewals
	Type	The type of call for tenders: open, by invitation, sole-source, etc.
	Status	The status: ongoing, opening results, completed.
	Title	The title of the contract
	Lots	The lots, if any
	Client	The client's name and its address and contact information.

Menu	Field	Description
Summary	Planned duration without options ¹	The base duration of the contract in months.
	Planned duration of options	The duration of the options in months.
	Publication date	The publication date of the call for tenders.
	Bid deadline	The closing date of the call for tenders
	Delivery region	The region in which the contract takes place.
Bid opening results	Bidder	The bidder's name.
	Amount submitted	The amount submitted by the bidder.
Contract award	Successful bidder	Whether the bidder was selected (there may be several).
	Contract amount	The amount of the awarded contract (often the same as the amount submitted, by definition).
Additional ex-penses	Expense	Amount and reason for the addition.
Final information	Contract end date	The end date of the contract.
	Number of options exercised	The number of options that were exercised.
	Total amount paid	The total amount paid to the bidder (may differ from the awarded amount for on-call contracts).
Correction history	Correction	Correction date and corrected fields.

4.4.2 Internal data

When the Commissionaires begin a tender response process, one of the first steps is creating a working folder in the cloud environment. In these folders, organized by year and date, there is at least a capture of the SEAO publication web page, as well as PDF files of the original contracts, specifications and other documents. A dashboard used to track the various publications is also updated and contains the general information of the contract, such as the client's name, the number of hours and the duration.

From the original contracts and the working documents that are sometimes available (for example, there are no working documents when the Commissionaires chose not to proceed to the bid), we built a database of the technical characteristics of the various contracts through manual collection from the documents. Since the scopes of the contracts' missions are very largely similar, we were able to define the important variables to consider; they are presented in Table 4.2.

¹Options are possibilities of extending contracts for a given period without republishing a call for tenders.

Table 4.2 Variables describing the technical characteristics of contracts

Variable	Description
Number	Unique client number of the contract
SEAO reference number	Number in SEAO; may be the same for some contract renewals
Title	The title of the call for tenders
Lot	The lot, if any
Client	The client's name
Date	The publication date
Client	The client's name
Planned duration	The base duration of the contract in months
Planned duration of options	The duration of the options in months
Result	The CCCQ's result
T ₂₀₂₀ , ..., T ₂₀₃₀	Number of hours over the decree periods (1 year)
P ₁ , ..., P ₁₃	Number of hours for each of the bonuses
Calls	Estimated number of calls (on-call contract)
Uniform	Binary: whether uniforms must be provided
Equipment	Binary: whether special equipment must be provided
Vehicle	Binary: whether vehicles must be provided
Region	The region where the contract is carried out
Spread	The geographic spread of the contract (in km) if there are several sites

The database, built from documents available since 2023, lists 307 contracts. The main challenge is that many open working folders contain only the original PDF files, and calls for tenders not studied by the CCCQ cannot be analyzed for lack of available documents. Moreover, it was difficult, if not impossible, to find certain key variables such as the cost of material resources; we therefore settle, at first, for considering global binary variables.

4.4.3 Cross-referencing the data

Once the internal and external databases have been built, the objective is to cross-reference them in order to link the technical characteristics of the contracts to the opening results of the calls for tenders. To do so, we use several key variables common to both databases, such as the number, the SEAO reference number, the publication date, the title, the lot and the client (only one as the primary key and the others as checks with tolerance), to link contracts and results.

We thus obtain a cross-referenced database of 272 contracts. In 97% (263) of cases, the opening results are available and, in 96% (262) of cases, the contract awards are available.

4.5 Conclusion

This chapter sets out the empirical context of the project. We described the tender response process currently in place, following the BPMN 2.0 formalism. This description reveals four structural weaknesses: weak integration of systematic indicators into decisions; difficult access to data; the absence of unbiased tracking of results; and prospecting that is still largely manual. Finally, we identified the data sources that can be used: the public SEAO platform, from which we extracted 1,960 publications between 2015 and 2026, and the internal database reconstructed from the partner's working folders since 2023 (307 characterized contracts), whose cross-referencing produces a unified database of 272 contracts. We will try to mitigate the identified weaknesses by introducing a tool and models based on the data presented. The next chapter presents the elements we developed during our research.

CHAPTER 5 PROPOSED TOOL AND ELEMENTS

In this chapter, we present the tool developed to support the tender response process. This tool integrates several models and sub-tools in order to address different needs or weaknesses of the process.

5.1 General presentation of the tool

The tool proposed in this chapter is a response to the weaknesses of the tender response process highlighted in the previous chapter. Rather than replacing the managers' work with an automated system, the approach adopted aims to design a decision-support system that fits into the existing steps of the AS-IS process and exploits the internal and external data available at each milestone. This stance preserves the manager's decision-making autonomy while providing context and quantification that were previously unavailable, particularly for complex decisions such as assessing the relevance of a call for tenders and determining the margin ratio.

The architectural choice is to favour a modular decomposition of the tool into five complementary elements, rather than a single solution. Several considerations justify this orientation. On the one hand, the problems identified in the previous chapter (insufficiently tooled prospecting, under-use of contractual PDF files, the lack of a systematic method for identifying comparable contracts, and the lack of a quantitative pricing tool) are of distinct technical natures (web retrieval, natural language processing, statistical learning, interactive visualization), each calling for specific methods. On the other hand, modularity allows progressive deployment and independent maintenance of the components, a property that is all the more relevant as the CCCQ's internal database is expected to grow with future bids, which will allow the models to be retrained without calling the overall architecture into question. This architecture is also a direct consequence of our DRM research methodology, in which iteration leads to the development or redevelopment of new elements after continuous evaluation of the other elements.

The articulation of the elements ultimately follows the chronology of the response process and is shown in Figure 5.1. Element 1, dedicated to prospecting, links the calls for tenders currently published on SEAO with their historical antecedents through entity resolution, in order to enrich the manager's upstream monitoring. Element 2 leverages the PDF files of the calls for tenders by extracting from them, by means of a hierarchical retrieval-augmented generation (RAG) architecture, quantitative variables related to material resource requirements, data that were not available in

structured form in the internal databases. Element 3 uses these variables, together with the technical characteristics listed in Table 4.2, to compute a similarity score between a contract under study and the CCCQ’s history, via a neural network trained with rankings made by experts. Element 4 exploits these contractual neighbourhoods and the market dynamics observed on SEO to formulate, using a Learning-to-Rank model, a margin ratio recommendation. Finally, the last element, Element 5, presents the outputs of the previous models in an interface intended for managers.

Technically, the whole is developed in Python to ensure interoperability between the components and to take advantage of the mature scientific ecosystem available in this language. The tool consumes data from both the SEO platform and the CCCQ’s internal SharePoint environment, then presents them to the manager following a dashboard logic, without automating sensitive strategic decisions. The next section details each of the elements, presenting, for each module, the data used, the method adopted and the first results obtained.

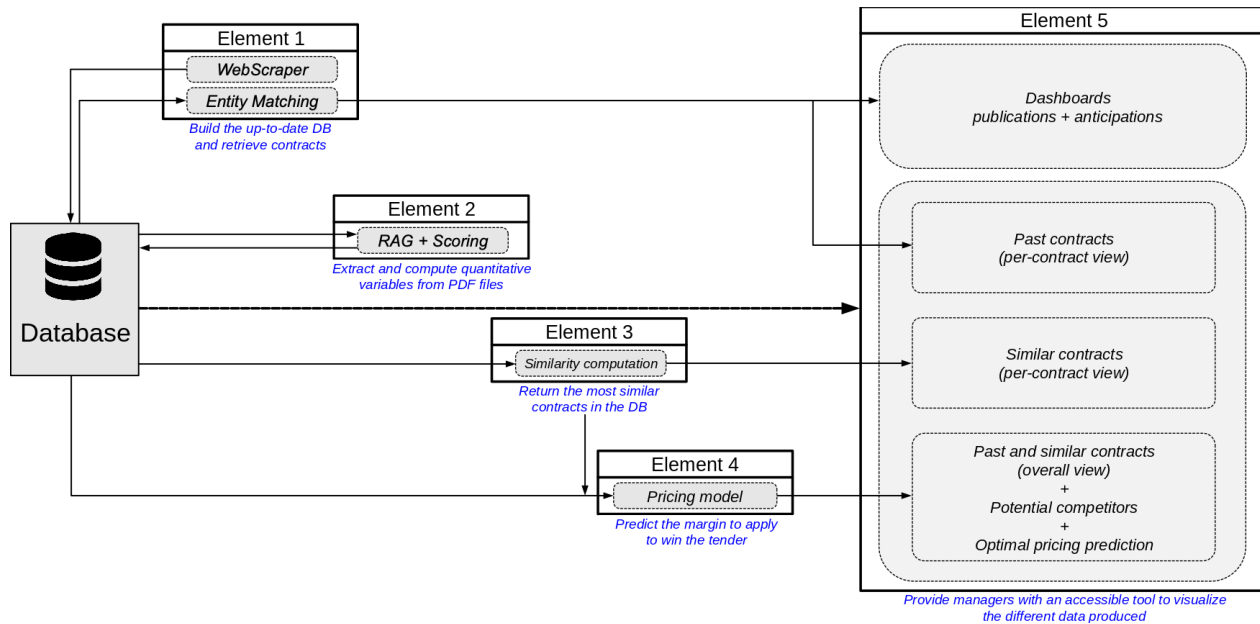


Figure 5.1 Articulation of the developed elements

5.2 Description of the integrated models and sub-tools

Our main tool is a combination of several other tools or models, which we will call elements, sometimes interdependent, addressing various problems and objectives. They are all developed in Python and allow the data to be used at several levels. These elements are detailed in Table 5.1.

Table 5.1 Developed elements

Element	Description	Inputs	Outputs	Means
1- Dashboard of publications and anticipations	Gives ongoing calls for tenders, with information and links if it is a renewal. Anticipates the publication of contract renewals	Web <i>scraping</i> of ongoing calls for tenders and SEAO historical database	Information and links to past calls for tenders if renewal.	<i>Web scraping</i> (Selenium); <i>Entity Resolution</i>
2- RAG for systematic data extraction	Quickly and systematically extracts data from the original PDF files	Original contract files in PDF format	Predefined information and computed scores for resource requirements.	<i>Hierarchical RAG; LLM</i>
3- Similarity computation model	Finds the contracts most similar to a contract under study	Defined variables representing the contracts	Similarity score between contracts	Distances; <i>Neural Networks; RLHF</i>
4- Pricing model	Gives an estimate of the winning margin ratio for lowest-price award	Market variables from the SEAO DB and contract characteristics	The winning margin ratio for lowest-price award	<i>LTR - XGBRanker</i>
5- Visualization tool	For a given contract, allows indicators to be visualized according to chosen variables	The outputs of the previous tools and models and the visualization choice variables	Visual indicators allowing the manager to explore the data	Streamlit

5.2.1 Element 1: Anticipation and historical mining

This first element aims to improve the process of prospecting for calls for tenders or contracts.

First, we automatically collect, via web scraping, the ongoing calls for tenders that are potentially of interest to us by filtering on titles and on UNSPSC codes (*United Nations Standard Products and Services Code (UNSPSC)*, 2026). For these ongoing calls for tenders, if they are renewals, we want to retrieve from the SEAO database the previous characteristics and results in order to display initial useful information to managers (price of the last contract, incumbent competitor, CCCQ's place in the call for tenders and link to the web page). At publication, we only have very general information for retrieving calls for tenders: the client's name, the contract title, the publication date and the reference number. Since these fields may change slightly for the same contracts (renewals), we use entity resolution methods (Rules + SequenceMatcher + Levenshtein + Soundex) to match calls for tenders, as illustrated in Figure 5.2.

Second, we seek to enable the anticipation of the publication of new calls for tenders or the prospecting of sole-source contracts coming to an end. To do so, we rely on the historical database

of contracts with their publication date and duration to determine possible republication dates. The same process as in our first point can then be applied to provide useful information.

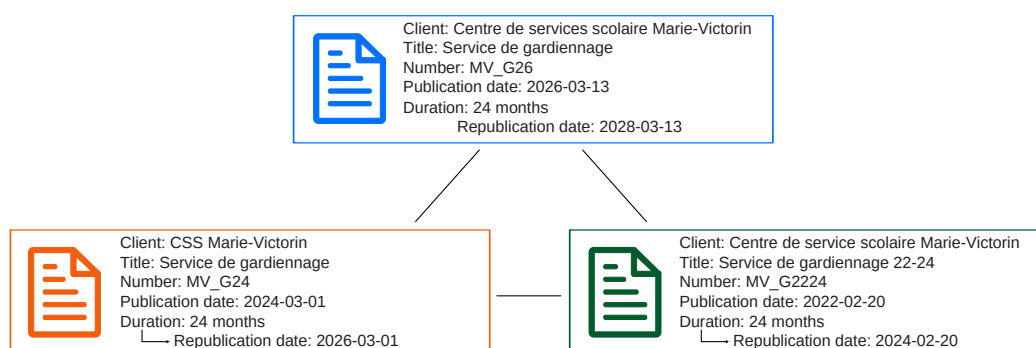


Figure 5.2 Example of contract linking by Entity Matching

These two points make it possible to set up a dashboard offering a slightly enriched view of the context of ongoing and upcoming calls for tenders. This contributes to improving managers' prospecting capabilities.

5.2.2 Element 2: Leveraging textual data

In this tool, we seek to leverage a large share of internal data, namely the contract files, technical specifications and other documents published by the client. These PDF files, often long (usually between 100 and 200 pages), contain valuable information both about the call for tenders and about the technical specifics of the contract. The main goal here is to extract variables quantifying the material resource requirements of the contracts, variables that were not available in a clean and rigorous form in the internal databases.

5.2.2.1 Retrieval-Augmented Generation

We use a RAG (*Retrieval-Augmented Generation*) structure, combining an *LLM* and the documents, to extract and estimate the resources required of the bidder. These resources, which we divide into three categories, concern requirements for uniforms, equipment and vehicles. Using an *LLM* for retrieval makes it possible to handle ambiguous cases where a resource is mentioned in the text but defined as being provided by the client rather than the bidder (Raini, 2024).

The structure we developed is illustrated in Figure 5.3.

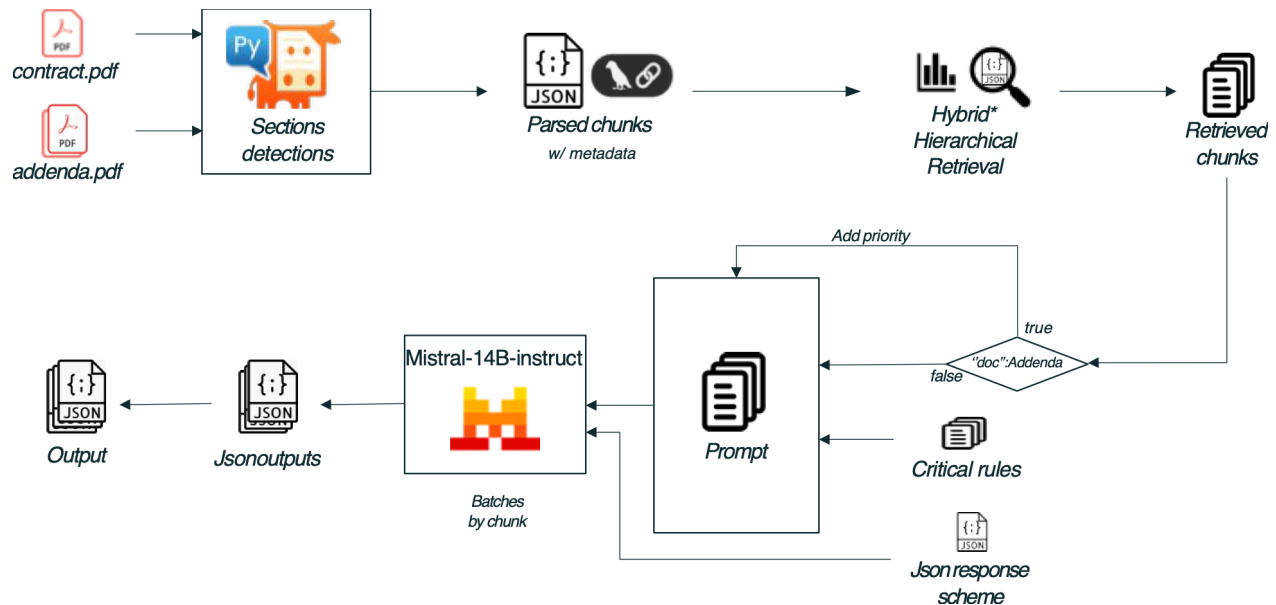


Figure 5.3 General structure of the RAG

First, we use the Python library *PyMuPDF* to read the PDF files. This library includes multiple vision tools, character recognition and models allowing it to segment text into paragraphs and label them (section title, text, list, footer, etc.) (PyMuPDF, 2024/2026). Since the contracts we process are highly structured, this gives us important information that we use afterwards.

With the retrieved information, we build the *chunks* (pieces of text) that will be used in the retrieval and generation phases. We use the hierarchical structure of the contract to create *chunks* representing the different sections at different levels (for example, “1.” is level 0, “1.1” is level 1 and “1.1.1” is level 2). This more logical segmentation of the text improves the efficiency of *retrieval* and of the method in general (Lu et al., 2025). A lower level then contains all the child *chunks* of higher levels. A pre-processing step, using purpose-built algorithms, recovers missing or mislabelled sections.

Since the vocabulary of security contracts is technical and specific, we favour embedding these chunks using sparse vectors (*sparse vectors*), augmented with domain-specific vocabulary (Iida & Okazaki, 2022). This approach is effective for retrieving the desired information and performs better than classical dense embeddings on specialized corpora (Mandikal & Mooney, 2024).

Next, we retrieve the relevant chunks. To do so, we rely on the sparse embedding vectors and on our query (the question asked) enriched with the domain vocabulary. We use the BM25 algorithm (Robertson & Zaragoza, 2009) coupled with the hierarchy to compare chunks; the complete method is presented in Appendix E. Figure 5.4 shows the improved performance of this method compared with classical methods in terms of information retrieval rate relative to chunk length, which directly affects the subsequent performance of the LLM in the generation phase (notably in

terms of hallucinations).

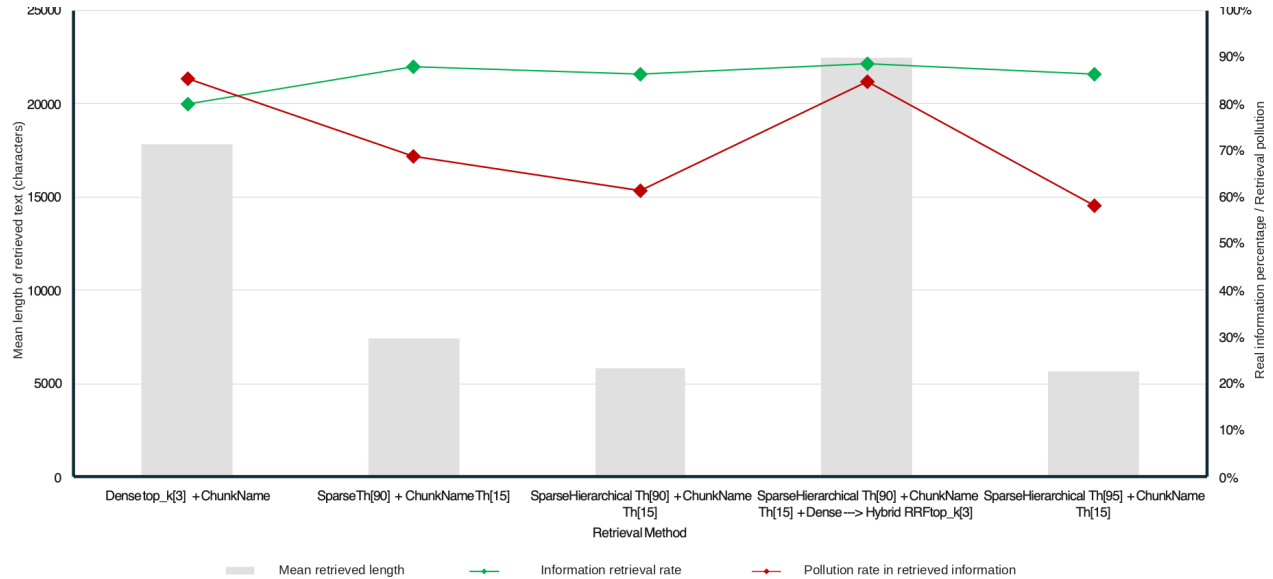


Figure 5.4 Comparison of *retrieval* methods

Once the *chunks* have been retrieved, we build the *prompt* passed to the *LLM*. If some chunks come from addenda, i.e., corrections or clarifications, the prompt specifies that this information should prevail in case of doubt. We add critical rules as well as the response schema the *LLM* must use. The language model we use is *Ministral-3-14B-instruct* (GGUF, Q5_K_M), a frugal model particularly well suited to our conditions. The results, in terms of retrieved elements (requirements for uniforms, equipment or vehicles), for 50 contracts are presented in Table 5.2.

Table 5.2 RAG extraction results

	Extraction	
	True	False
True	263 (88%)	5 (1%)
False	41 (13%)	.

We thus observe very satisfactory performance, with a precision of 87% and a recall of 98% on the elements identified in 50 contracts.

5.2.2.2 Quantitative assessment of resource requirements

After extracting a list of resource requirements, we quantify the requirements in each category (uniforms, equipment and vehicles). To do so, we build a fairly exhaustive dictionary of costs (in

dollars) for recurring identified requirements. The goal here is not to quantify the exact financial cost but to build a basis for comparison between contracts. From the extracted list of resources and using the NLP library *spaCy* (Stargazers · Explosion/spaCy, 2026), each resource is assigned its cost, taking into account exact or partial matches, in which case the costs are distributed. The sum gives the score corresponding to the resource requirements.

We can evaluate our method for the uniforms category. Indeed, the distribution of the scores matches the cost charged by the Commissionaires for different types of uniforms, as illustrated in Figure 5.5.

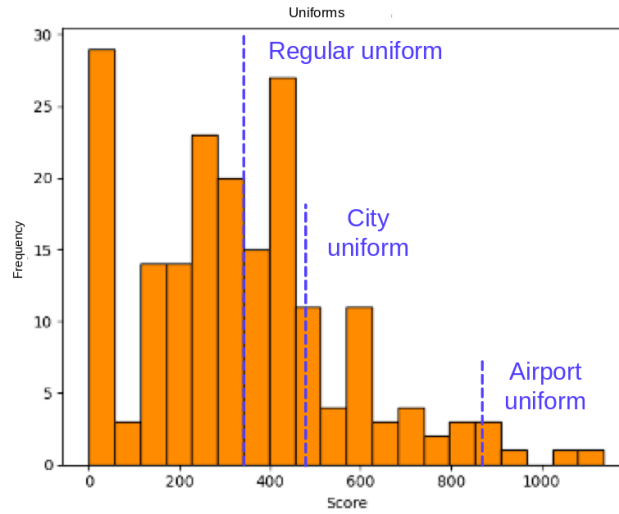


Figure 5.5 Distribution of uniform scores (orange) compared with the prices charged by the CCCQ (blue)

5.2.3 Element 3: Computing contract similarity

In this element, we develop a way of comparing contracts in order to retrieve the most similar contracts. The variables we consider are listed in Table 5.3. All continuous variables were normalized using min-max scaling.

Table 5.3 Variables used for contract similarity

Variables	Types	Comments
Region	category	16 regions
Client type	category	8 types
Client name	string	
Duration	float	In months
Publication date	date	
Price	float	Decree price
Calls	float	Number of calls per year
Hours	float	Number of hours per year
Vehicles	float	RAG score for vehicle requirements
Equipment	vector[float]	RAG scores for uniform and equipment requirements
Nature	vector[float]	Vectors of bonus hours

First, we define distances between the variables of two contracts:

- Region: we define a distance matrix based on characteristics of the regions (urbanization, population, area);
- Client type: we define a distance matrix according to the clients' characteristics; for example, "CEGEPS" and "School service centres" are close;
- Client name: sequence comparison;
- Duration, Hours, Price, Equipment: simple Euclidean distance;
- Publication date: computation of the delta followed by logarithmic decay;
- Nature: combination of binary and weighted Jaccard distances in order to capture matches in terms of the bonuses present and their quantity; and
- Vehicles, Calls: Euclidean distance with a penalty in case of binary non-equality (Eq 5.1). These variables strongly influence the type of contract (presence of vehicles or on-call hours).

$$d(v1, v2) = \begin{cases} 0 & \text{if } v1 = v2 = 0 \\ |v1 - v2| & \text{if } v1 \neq 0 \text{ and } v2 \neq 0 \\ \lambda |v1 - v2| & \text{otherwise} \end{cases} \quad (5.1)$$

All the distances (except in the case of penalties) give a result between 0 and 1; our goal is now to combine them to obtain a distance representing the overall similarity between two contracts. The method we chose combines a naive approach, assigning an initial weight to each of the distances, with a machine learning approach to fine-tune these weights.

We therefore first assigned unit weights to each of the distances and trained a simple neural network model (3,2). At this stage, training is very fast. Next, we build a ranking database by taking one contract and 3 others among the 15 most similar, and asking an employee to rank them from most to least similar. This database, made up of about 300 iterations, is used to retrain the network through reward-based learning to adjust the relevant weights, using the Bradley-Terry loss and an implementation of the RankNet algorithm (Burges et al., 2005) to compare rankings.

Figure 5.6 shows the model's performance before and after reinforcement. The share of exact triplets increases significantly. Likewise, we observe an increase in the Spearman rho correlation coefficient, which confirms that the model has integrated the human ranking reasoning.

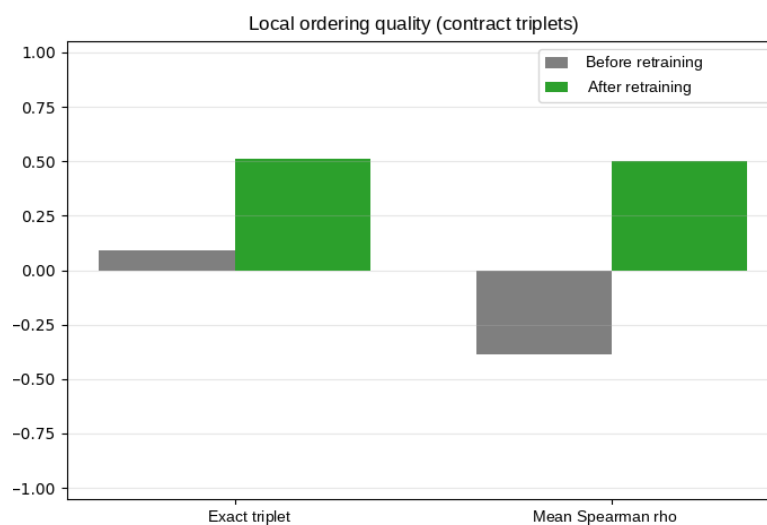


Figure 5.6 Similarity after retraining

5.2.4 Element 4: Pricing model

In this model, we rely on the tools developed previously to build, from historical data, a model that predicts the margin to apply in order to beat the other bidders in a lowest-bidder call for tenders.

The goal of our model is to consider both the environment and competitive context of the current market and the technical characteristics specific to the contracts.

5.2.4.1 Contract characteristics

In our model, to take into account the technical characteristics of the contracts, we use the similarity model developed previously. We thus extract the most similar contracts in order to use the results of the associated calls for tenders.

The variables we retain are the winning margin ratios and the dates of the calls for tenders of these k similar contracts. However, these calls for tenders may have taken place at more or less distant dates, in a market that may no longer resemble today's. This is why we must take into account the evolution of the dynamics of the market in which we operate.

5.2.4.2 Market dynamics

A first way of taking market dynamics into account consists of updating the winning margin ratios of the similar contracts. To do so, we rely on the evolution of winning margin ratios over time. For a call for tenders published at date t , we consider how the winning ratio stood relative to the market at that time to infer the ratio around which it should stand relative to the current average M_T , as illustrated in Figure 5.7. Simply put, we compute the updated ratio R_T by simple proportionality (Eq 5.2).

$$R_T = \frac{R_{t-1} \cdot M_T}{M_{t-1}} \quad (5.2)$$

We thus update the winning margin ratios of the similar contracts.

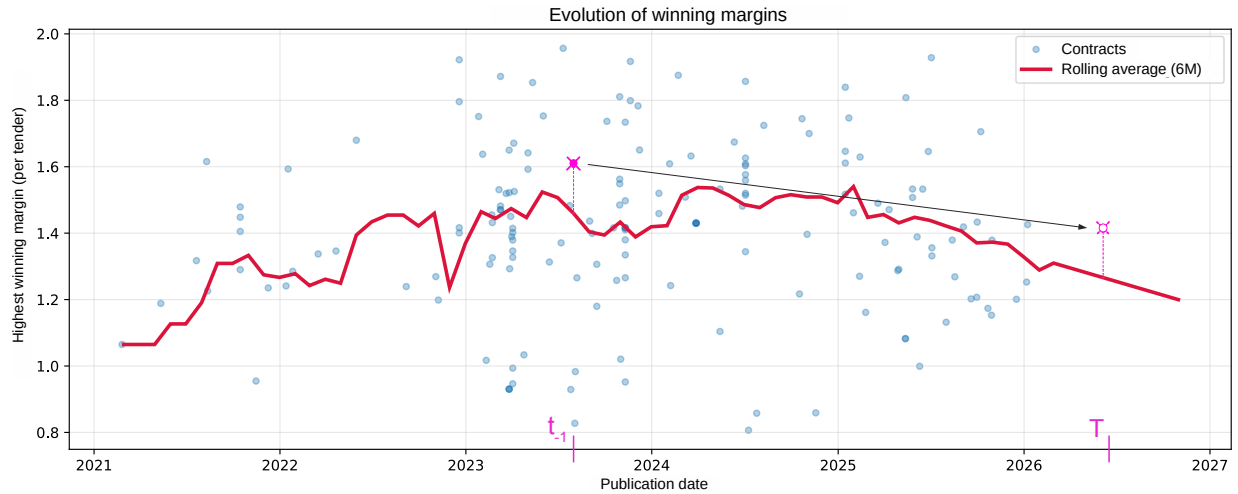


Figure 5.7 Market evolution – Example of updating at date T for a contract published at date t_{-1}

The second way of taking market dynamics into account consists of looking up the latest calls for tenders and their results in order to compute an average of the current winning ratios.

The two inputs we will later use in our model are therefore:

- The winning ratios of similar contracts, updated to the current market; and
- The winning ratios of the latest contracts.

5.2.4.3 Models

The pricing problem in calls for tenders has been studied for some time. The first works, by Friedman and Gates, approach the problem from a statistical point of view by studying the distribution of the margins applied by the different competitors to establish probability density functions of winning (Friedman, 1956; Gates, 1967). The study of these statistical models continued to evolve, notably with Carr (1982), Skitmore (2002, 2004, 2014) and colleagues (Skitmore et al., 2007; Ballesteros-Pérez & Skitmore, 2017), in order to better model the distribution of competing bids and refine the estimate of the conditional probability of winning at a given margin. However, these models rely on strong assumptions (independence of bidders, market stationarity, and the availability of a large number of comparable bids) that are rarely verified in practice and limit their ability to integrate the technical context specific to each contract.

To overcome these limitations, the literature gradually turned to approaches from operations research, then from machine learning. Artificial neural networks (Christodoulou, 2010; Moselhi & Hegazy, 1993) then demonstrated a better ability to model these complex relationships, at the cost of a large need for training data and low interpretability, constraints that are particularly penalizing in our tendering context where the usable history is limited. Case-based reasoning approaches have also been explored (Chua et al., 2001) to explicitly exploit similarity with past contracts, but they remain sensitive to the quality of the similarity measure and do not naturally integrate the temporal dynamics of the market. More recently, ensemble methods based on decision trees, random forests and gradient boosting have emerged as an attractive compromise between performance, robustness to overfitting and the ability to work with moderately sized datasets (Chen & Guestrin, 2016).

5.2.4.4 Learning to Rank model

In our case, the database available to us contains about 120 usable calls for tenders, which rules out deep architectures and weakens the classical formulation of the problem as a regression on the absolute margin. We therefore turn to a Learning to Rank (LTR) approach based on XGBoost (Burges et al., 2005). Reformulating the problem, no longer as predicting a margin value but as a relative ranking of potential bids against one another, has several advantages: it makes better use of the pairwise comparisons made possible by the similar contracts identified in the previous step; it is less sensitive to the absolute variations in margins induced by market dynamics, these being already partly neutralized by the updating described in Equation (5.2); and it allows a limited volume of data to be used more effectively. XGBoost also brings the ability to model non-linear interactions between the contract characteristics, the updated ratios from similar contracts and the ratios of the latest calls for tenders won, while maintaining relative interpretability through the

analysis of feature importances.

The model, described in the diagram of Figure 5.8, is based on a grid of plausible candidate margins (ratios) (from 1.05 to 3.00) as well as their position relative to the margins of the latest contracts or of similar contracts; these are the nine features of the ranking model. We then identify the relevance of a bid, the highest possible winning bid being the most relevant bid since it allows the contract to be won while applying the maximum margin; non-winning bids are by definition not relevant.

To evaluate our model, we rely on a reference base of the Commissionaires. This database contains 54 calls for tenders in which the bidders participated. We compute the Commissionaires' reference performance on the last 20 calls for tenders, which constitute our test set. For our model, for each of the contracts in the test set, we train our model on all the previous contracts (chronological Leave-One-Out), which corresponds to the real conditions of potential use of the model.

The performance indices we use are listed in Table 5.4.

Table 5.4 Performance indices of the pricing model

Performance index	Formula (for N calls for tenders)
Rank Score (RS)	$\sum_i^N \frac{1}{r_i}$ with r_i the rank (5.3)
Mean Rank (MR)	$\frac{1}{N} \sum_i^N r_i$ with r_i the rank (5.4)
Win rate (WR)	$\frac{1}{N} \sum_i^N (r_i = 1)$ with r_i the rank (5.5)
Mean Distance (MD)	$\frac{1}{N} \sum_i^N m_i - w_i $ with m_i the predicted margin and w_i the winning margin (5.6)
Cumulative Distance (CD)	$\sum_i^N m_i - w_i $ with m_i the predicted margin and w_i the winning margin (5.7)
Dollar Gains (DG)	$\sum_{r_i=1}^N (m_i - 1) p_i$ with m_i the predicted margin and p_i the contract price (5.8)

We compare the results of the performance indices of several models and methods, on this same test set, with the Commissionaires' reference results in Table 5.5.

Table 5.5 Results of the different models

Model	RS↑	MR↓	WR↑	MD↓	CD↓	DG↑	Variables
CCCQ reference	0.26	6.20	15%	0.362	7.25	\$2.17 M	/
Gates-Friedman ¹	0.18	8.80	17%	0.412	8.29	\$0.72 M	Distribution of historical margins
Means	0.35	6.50	25%	0.318	6.36	\$1.50 M	Similar and recent contracts, no market adjustment
Regression	0.50	4.15	40%	0.250	4.99	\$1.59 M	Similar and recent contracts, market-adjusted averages
LTR	0.49	3.55	40%	0.280	5.58	\$3.52 M	Similar and recent contracts, market-adjusted variables

The performance analysis demonstrates the superiority of the selected approach, Learning to Rank (LTR), which manages to maximize both the win rate and profitability. While the reference strategy (CCCQ) and the historical statistical methods struggle to reach win rates (WR) between 15% and 25%, using Machine Learning (Regression and LTR) achieves a win rate of 40%. However, a difference separates Regression from LTR: although Regression is very good at “aiming right” and minimizing the absolute deviation from the winning margin (MD of 0.250), it generates low financial gains (\$1.59 M). Indeed, the regression, based solely on the averages of the winning margins of the latest contracts and of similar contracts, uses a symmetric error that pushes it to underprice bids in order to secure the win at the expense of profit. Conversely, the LTR model stands out as the best model, with good financial gains (\$3.52 M) and the best mean rank (MR of 3.55). By evaluating scenarios on a grid of candidates (from 1.05 to 3.00) rather than predicting an absolute value, LTR makes full use of the small volume of data and captures the asymmetric nature of the auction: it does not simply look for the lowest price, but precisely identifies the highest possible bid that beats the competitors, thereby directly maximizing the company’s profitability.

¹We choose Gates’ model with a win probability threshold of 80%, a threshold chosen after carrying out a sensitivity analysis.

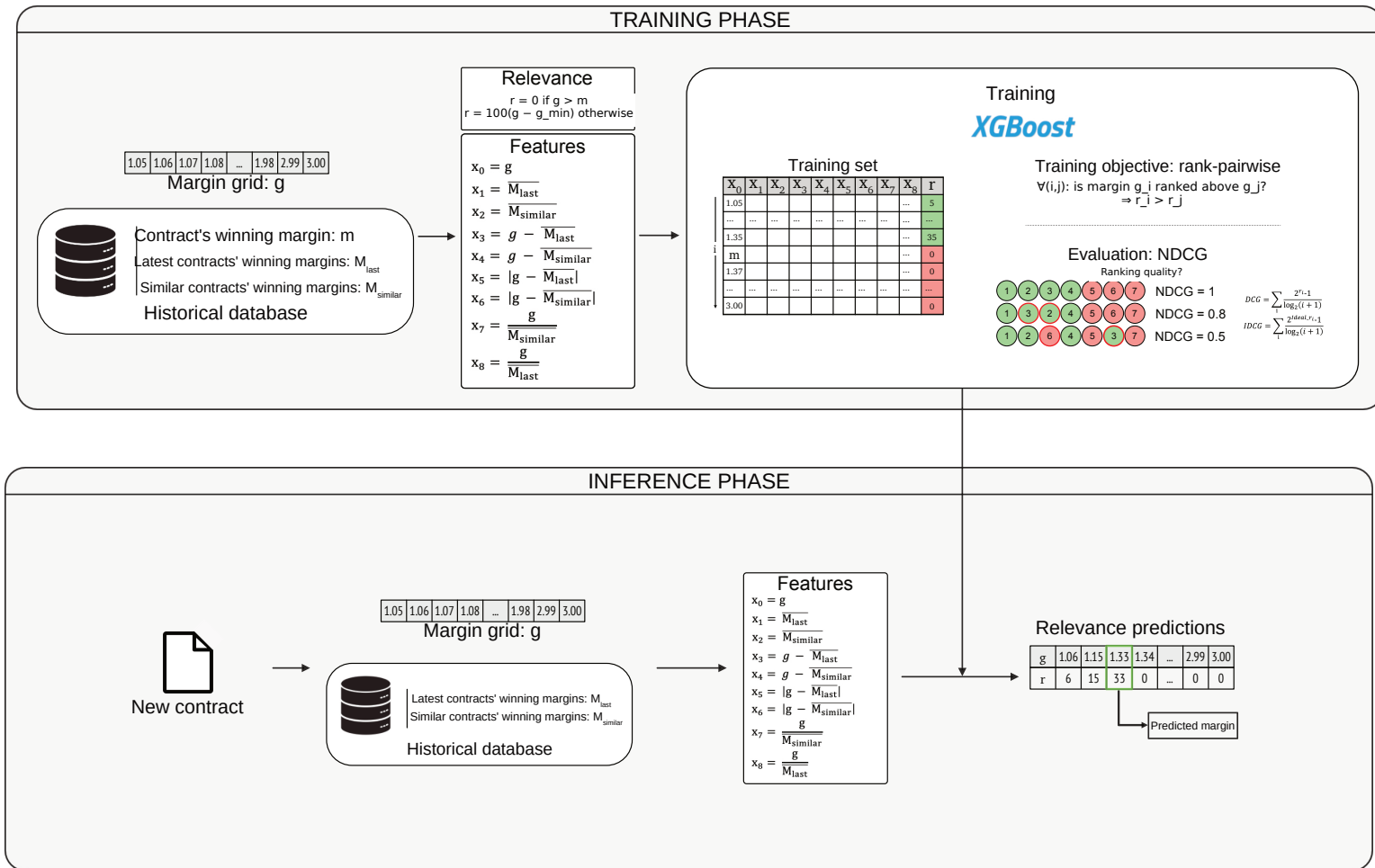


Figure 5.8 Diagram of the pricing model

We have thus developed a model capable of predicting a winning and profitable margin using the characteristics of the contracts, measured by similarity, as well as the characteristics and dynamics of the market.

5.2.5 Element 5: Visualization tool

All the elements developed previously must be able to fit into the tender response process. These data are therefore integrated into a tool that managers can consult and interact with. This tool is designed as an application, so that it can potentially be integrated into an internal environment in the future.

A first part, or page, of this tool is Element 1. This page (Figure 5.9) is useful during the upstream phase of the process: prospecting for new contracts. It gives the manager a quick overview of ongoing and upcoming calls for tenders and helps quickly identify those with the most potential.

This page gives access to ongoing contracts (first table) and to upcoming renewals (second table). An ongoing notice can be selected. The “Analyze this contract” button takes the user directly to the analysis, pre-filling the general information fields of the analysis page.

The second part, the analysis page (Figure 5.10), makes it possible to select a call for tenders and enter its technical characteristics. The tools and models developed in our research project are then used to display the various indicators and exploration menus useful to decision-makers in their bidding decision and in building their financial offer.

A first example of use is the “matching past contracts” sub-menu, which retrieves the previous calls for tenders of the contract whose renewal is being studied. It is then possible to consult the corresponding page of the call for tenders on the SEAO website via the retrieved links, but also to display the results, the bids and the margins applied directly, as well as the computation of the updated margins. The raw data available can also be displayed, in JSON format.

Second, similar contracts are retrieved and displayed in the same way.

Finally, a general analysis section is available to study the context of the contract. Charts display the margins (actual and updated) for the latest contracts, similar contracts and past contracts. The potential competitors are then displayed, along with their participation rate in the calls for tenders issued by the client, their win rate with this client when participating, the last five margins applied, and the competitor's general pricing trend over the previous year. This allows us to assess how threatening a competitor is according to its affinity with the client and its current aggressiveness. The last indicators displayed are the average margins for the latest contracts, similar contracts and past contracts, as well as the margin recommendation computed by the pricing model developed.

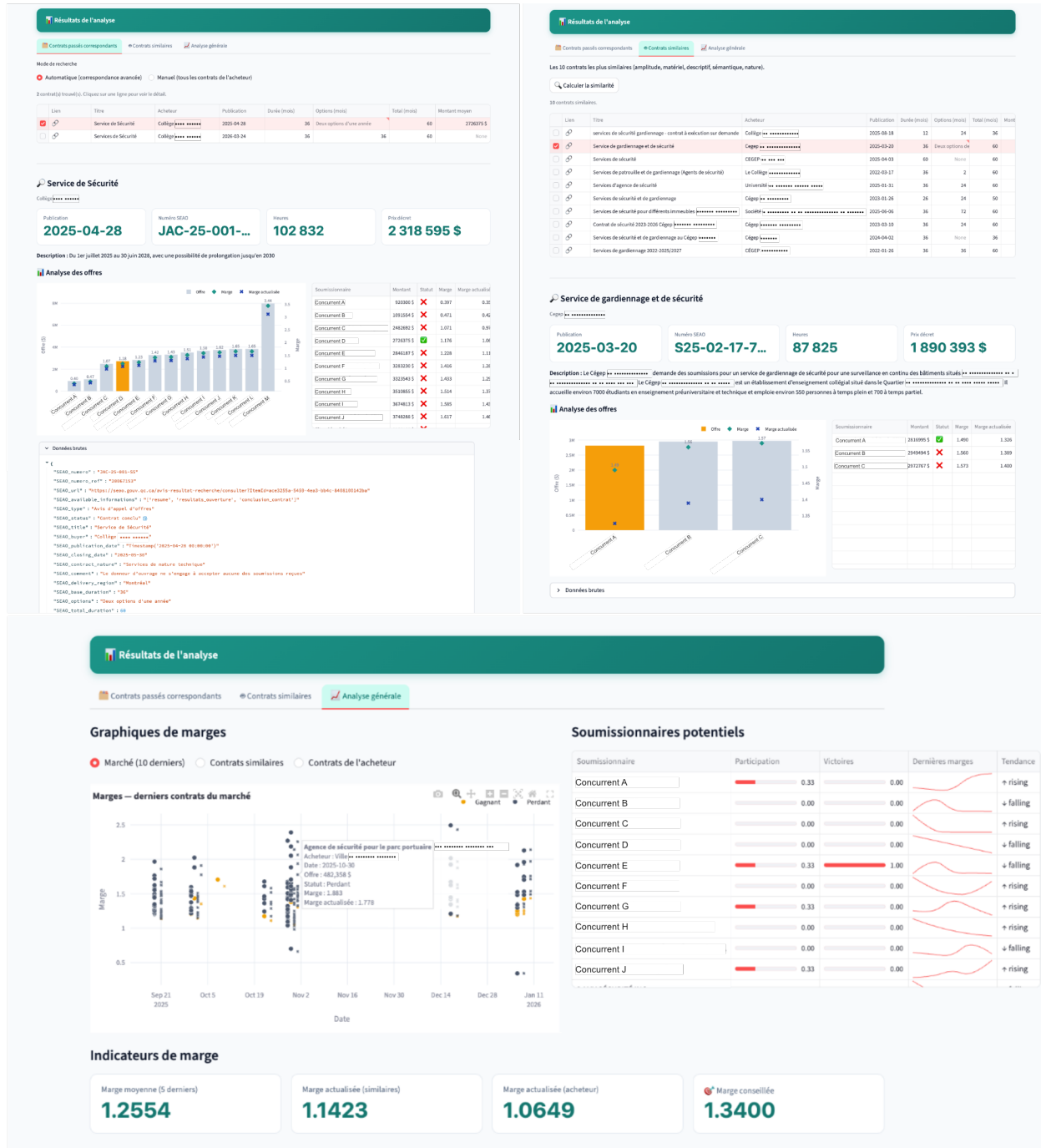


Figure 5.10 Second page of the tool (Analysis)

5.3 Integration into the process

Integrating our tool into the process brings data into the targeted process at our partner. To highlight the changes made to the process, we use a colour code to show their nature. Modified activities are

shown in yellow, new activities in green and activities that are only moved chronologically in blue (Mosser, 2020).

The first part of the tool completely transforms the stance taken when prospecting for new calls for tenders. Rather than waiting for email alerts configured on the SEAO platform or consulting it from time to time, all ongoing calls for tenders are quickly accessible visually. In addition, key information such as the likely size (price) of the call for tenders, the incumbent competitor and the CCCQ's position in the last call for tenders in the case of a renewal makes it possible to quickly see the high-impact calls for tenders. Finally, an anticipatory stance is made possible by showing potential future calls for tenders or sole-source contracts coming to an end. This can help managers better prioritize their response potential, taking into account the current and future context of the market for tender openings. A global context is thus provided to support the decision to bid.

The second part of the tool transforms the process of analyzing and building the bid. It now includes data that previously were not considered in the process for lack of availability or of a tool allowing their use. During the bid analysis phase illustrated in Figure 5.11, a step is now added to systematically save contract-related data, data that are essential for building a database linking contracts to the results of calls for tenders. It is also these data that subsequently enable in-depth analysis and provide crucial indicators to managers. Then, during the phase of computing the financial offer illustrated in Figure 5.12, i.e., computing the margin ratio to apply, the manager can now rely on historical data and on models to establish, in a more quantitative way, the optimal strategy to adopt. These indicators support the decision by providing the historical context, the technical context and the current market dynamics.

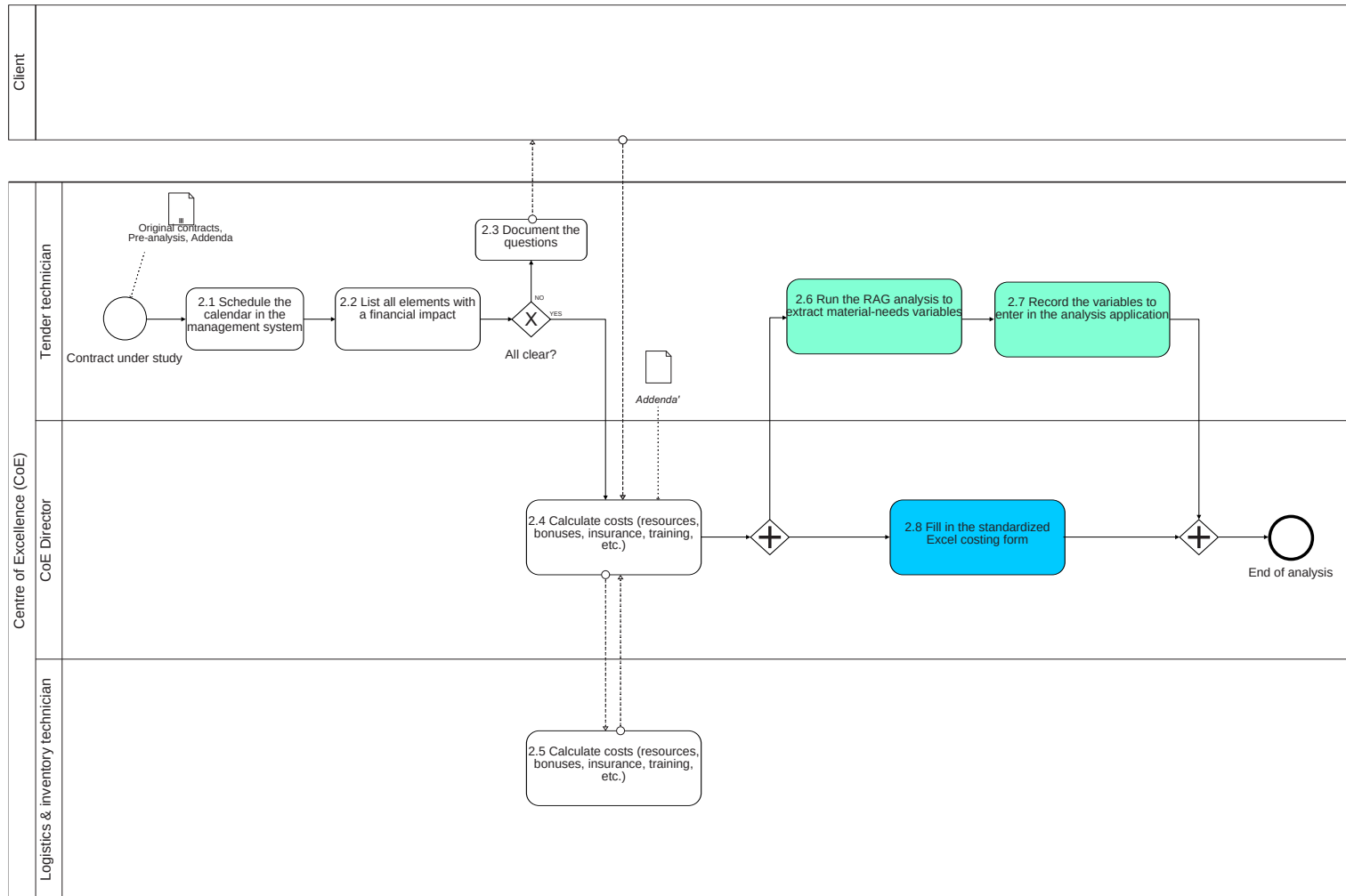


Figure 5.11 Analysis process (TO-BE)

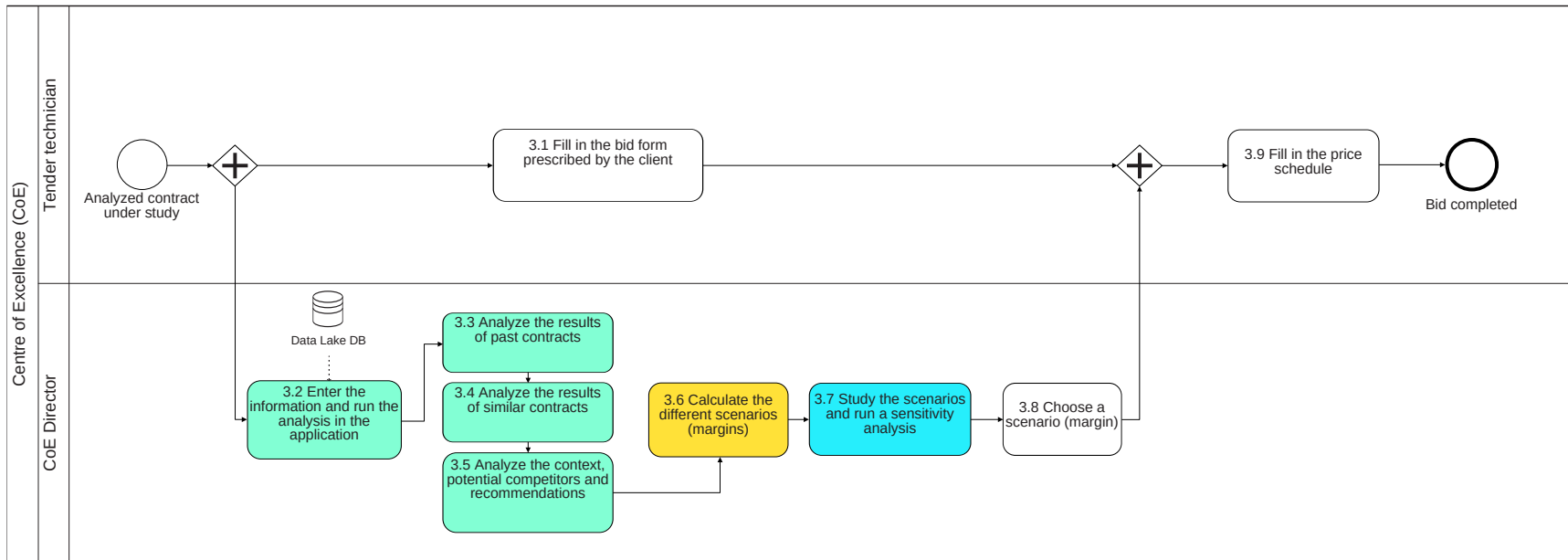


Figure 5.12 Scenario analysis process (TO-BE)

5.4 Deployment

The tool is being deployed in our partner's environment in stages, and the first work under way concerns Element 1 (and part of Element 5). Element 1, which is the first page of the graphical tool (Element 5) dedicated to prospecting and anticipating renewals, is being deployed within the Commissionaires' Azure infrastructure.

The architecture adopted is built around two main components. On the one hand, a data storage and orchestration layer, set up in the partner's data lake (Azure Data Lake Storage Gen2), centralizes all the data needed by the tool. The data on the opening results of calls for tenders are updated weekly, in line with the publication rhythm observed on the SEAO platform, while ongoing notices are retrieved daily to ensure an up-to-date view of market opportunities.

On the other hand, the application itself is containerized with Docker and hosted in the partner's Azure environment. This architectural choice meets several objectives. It allows the maintenance and evolution of the tool to follow a DevOps approach, facilitating the continuous integration and deployment of upcoming versions. It also ensures uniform availability of the application for all users of the environment, regardless of their workstation, and provides clean isolation of the technical dependencies needed to run the Python components developed.

This first deployment thus constitutes a proof of concept and a foundation on which the next elements of the tool can progressively be put into production.

5.5 Conclusion

In this chapter, we described the tool and the elements we propose for integrating data into the tender response process studied at our partner. We also examined how this tool could be integrated at our partner and how it changes the processes in place. In the next chapter, we test our tool and its elements by applying them to potential calls for tenders available in our historical database. This will allow us to compare our recommendations and indicators with the actual decisions made in order to evaluate the usefulness of our tool.

CHAPTER 6 EXPERIMENTS

The objective of this chapter is to evaluate the relevance and performance of our tool by using it under real past conditions. We use our historical database to evaluate our tool at different points in time, reflecting diverse contexts and situations. We can thus identify the weaknesses, strengths and potential improvements needed.

6.1 Calls for tenders used

For our experiments, we use the historical data available in our database. These data contain calls for tenders to which the CCCQ paid attention, in which case we can carry out a complete analysis, but also calls for tenders not considered by the CCCQ, in which case we cannot go further than studying our prospecting tool. When studying a published call for tenders, we therefore consider all the data available before the publication date minus two months, the average time for the results of a call for tenders to be posted online. This period indeed constitutes a zone of results publication that is normally invisible, since the results of the contracts published in this zone cannot have been published before the date of our study. This scheme is illustrated in Figure 6.1.

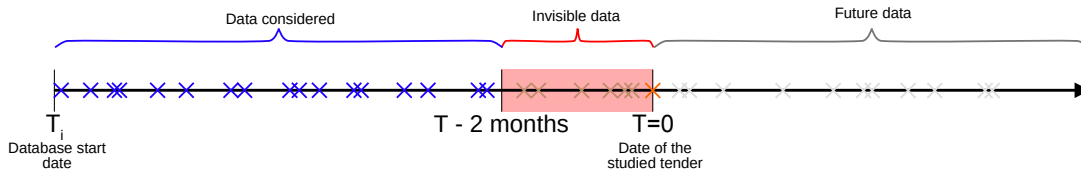


Figure 6.1 Diagram of the data used in an experiment

6.2 Prospecting and anticipation

The purpose of this section is to evaluate our prospecting tool (the first page of our tool); to do so, we check that the information computed and displayed is correct. In this section, the technical data of the contracts are not needed for the fields we seek to verify, so we do not depend on the CCCQ's internal database and can therefore consider more calls for tenders from the SEAO database. We thus consider all the calls for tenders published in 2025, i.e., 179 contracts.

First, we seek to verify the retrieval of previous contracts in the case of renewals. We obtain 60

contracts for which our *Entity Matching* algorithm found past contracts. The concordance is 92% (101 out of 110); only a few contracts matched by the algorithm have no direct link. Table 6.1 presents some examples found.

Table 6.1 Matching results by *Entity Matching*

Title	Date	Price
Service d'une agence de sécurité pour la surveillance de l'unité sanitaire – Ville de Québec	Feb-25	
Service d'une agence de sécurité pour la surveillance de l'unité sanitaire mobile - Place du Carré Lépine – Ville de Québec	Jul-23	
Service d'une agence de sécurité pour la surveillance de l'unité sanitaire mobile - Place du carré-Lépine – Ville de Québec	May-24	
Service d'agence de sécurité Centre Jean-Claude-Malépart 2025-2026 – Ville de Montréal - Service de l'approvisionnement	Apr-25	350,666
Services d'agence de sécurité au Centre Jean-Claude Malépart – Ville de Montréal - Service de l'approvisionnement	Mar-24	381,671
Service de sécurité pour l'immeuble sis au 303, rue Notre-Dame Est, Montréal, Complexe CDL – Société d'habitation et de développement	Apr-25	
Services de sécurité pour l'immeuble situé au 303, rue Notre-Dame Est à Montréal – Société d'habitation et de développement	Mar-23	
Services de sécurité pour l'immeuble situé au 303, rue Notre-Dame Est (Complexe CDL) – Société d'habitation et de développement	Mar-22	
Service de sécurité pour un bât., divers parcs et événements de la ville – Ville de Salaberry-de-Valleyfield	Apr-25	96,834
Service de sécurité pour un bâtiment, divers parcs et événements de la ville – Ville de Salaberry-de-Valleyfield	Feb-25	
Fourniture de services de sécurité pour la Ville de Longueuil – Ville de Longueuil	May-25	580,28
FOURNITURE DE SERVICES DE SÉCURITÉ POUR LA VILLE DE LONGUEUIL – Ville de Longueuil	Apr-22	591,956
Service d'une agence de sécurité Bibliothèque Paul-Aimé-Paiement (2019-289) – Ville de Québec	Jun-25	
Service d'une agence de sécurité (gardien) - bibliothèque Paul-Aimé-Paiement. – Ville de Québec	May-24	
Services de sécurité de Blainville en fête et l'OSM dans les parcs, édition 2025 – Ville de Blainville	Aug-25	
Services de sécurité de Blainville en fête édition 2024 – Ville de Blainville	Jul-24	
Fourniture sur demande de service d'agents de sécurité pour le Service des loisirs et de la culture – Ville de saint-georges	Aug-25	58,723
Fourniture sur demande de service d'agents de sécurité pour le Service des loisirs et de la culture – Ville de saint-georges	Sep-23	73,239
Fourniture sur demande de service d'agents de sécurité pour le Service des loisirs et de la culture – Ville de saint-georges	Nov-24	77,838
<i>Fourniture sur demande de service d'agents de sécurité pour le centre de ski – Ville de saint-georges</i>	<i>Nov-23</i>	<i>26,869</i>
Service de gardiennages dans divers lieux de l'arrondissement Lachine – Ville de Montréal - Arrondissement de Lachine	Dec-25	375,611
Service de gardiennage pour différentes bâtisses de l'arrondissement Lachine – Ville de Montréal - Arrondissement de Lachine	Dec-22	319,48

Second, we check the anticipations. In 2025, 177 contracts had a possible renewal date (end of the base period or end of the base period + options). Of these 177 contracts, 21% (38) were indeed republished as new calls for tenders at similar periods in 2025.

6.3 Analyses

This section aims to evaluate the indicators provided by our tool as well as the pricing model when analyzing a call for tenders.

6.3.1 Evaluation of the general pricing model

Using the data scheme described previously, we study the calls for tenders issued over a one-year period, from February 2023 to March 2024. These contracts represent all those present in our database for this period for which we have sufficient data to allow prediction by our model. In other words, the CCCQ paid at least some attention to them, which led to the original contract files being downloaded and archived.

For these 57 calls for tenders studied, the number of training data points for the model ranges from 31 for the oldest call for tenders to 117 for the most recent. The results are reported in Table 6.2.

It should be noted that a margin predicted by the model may in practice be irrelevant; indeed, below a certain threshold (~ 0.3), the contract is not economically viable for the CCCQ. This is directly due to the calculation method: the margin is in fact the margin ratio (Eq 4.1), which does not take all fixed costs into account. Profits are therefore calculated by subtracting this threshold from the applied margin (Eq 6.1).

$$Profit = Minimum\ price\ regulated\ by\ the\ decree\ (Margin\ ratio - (1 + threshold)) \quad (6.1)$$

These margins are coloured orange in the table and, even if their quantitative value is not relevant, they can signal that the call for tenders is too competitive to be viable for the CCCQ.

Of the 57 calls for tenders, the model wins 59% (33 first places). However, in 48% of these (16 calls for tenders), the margin is too low to be profitable. If a participation threshold is set to consider only profitable margins, the model achieves a win rate of 25% (15 first places), with a profit of about \$9 M. Of the 40 participations then considered, the model obtains 15 first places, 7 second places and 6 third places, placing it in the top 3 in 70% of cases.

Table 6.2 Model results for the calls for tenders of the 2023–2024 period

Call for tenders	Date ^a	N_train	Model (margin)	Last winner (margin)	Price	Rank	Model profit
1000529_1695032	f-23	31	1.17	2.00	607,852	1	-79,021
87390_1697297	m-23	32	1.70	1.53	115,545	3	
CISSSO-2324-011-S_1701134_Lot 01	m-23	33	1.60	1.48	7,428,380	5	
CISSSO-2324-011-S_1701134_Lot 02	m-23	33	1.60	1.47	1,035,180	3	
APL00013178_1700907	m-23	33	1.73	1.87	2,586,589	1	1,112,233
40366100_1704866	m-23	37	1.21	1.52	1,676,282	1	-150,865

Call for tenders	Date ^a	N_train	Model (margin)	Last winner (margin)	Price	Rank	Model profit
SE-2023-5632-001-AOP_1707085_Lot 08	m-23	38	1.29	1.52	76,856	1	-769
DG-2023-001_1707077	m-23	38	1.29	1.65	188,907	1	-1,889
AOP20230329NH-SM-SEC-ETE_1707964	m-23	40	1.49	1.29	195,898	3	
CL_222300-7_1708984	m-23	41	1.50	1.45	1,322,040	2	
44533253_1709697	a-23	42	1.46	1.39	1,261,991	2	
44533278 et autres_1710129_Lot 04	a-23	43	1.18	1.40	1,106,255	1	-132,751
44533278 et autres_1710129_Lot 03	a-23	43	1.19	1.33	381,786	1	-41,996
44533278 et autres_1710129_Lot 06	a-23	43	1.28	1.38	147,914	1	-2,958
44533278 et autres_1710129_Lot 07	a-23	43	1.19	1.35	185,203	1	-20,372
DR23-PUB-061_1711014	a-23	48	1.17	1.67	195,196	1	-25,375
23-6108_1711753	a-23	49	1.51	1.53	1,488,735	1	312,634
367100113_1717268	a-23	50	1.43	1.03	41,968	4	
87891_1720434	m-23	51	1.30	1.59	171,949	1	
87890_1719422	m-23	51	1.28	1.64	310,463	1	-6,209
87919_1721393	m-23	53	1.41	1.85	1,586,999	1	174,570
SER-SECURITE 2023_1722747	m-23	54	1.34	3.21	485,280	1	19,411
88123_1730043	j-23	55	1.44	1.75	288,793	1	40,431
6000025686_1735234	j-23	56	1.55	1.31	360,721	1	90,180
2022/23-226_1733867	j-23	57	1.48	2.44	137,425	1	24,737
CT-4485_1742771	j-23	58	1.55	1.37	2,518,464	3	
88409_1743089	j-23	59	1.43	1.96	573,256	1	74,523
2023-7553-50-01_1692797_Lot 3	j-23	60	1.63	3.80	3,335,645	1	1,100,763
2023-7553-50-01_1692797_Lot 2	j-23	60	1.63	3.81	1,566,903	1	517,078
AO-2023-01_1747549	j-23	64	1.26	1.48	921,532	1	-36,861
CSI2324-021_1749120	j-23	65	1.41	2.06	756,017	1	83,162
MACM 2023 00002_1749868	j-23	66	1.86	1.61	1,891,858	3	
44534900_1751474	a-23	67	1.36	1.27	321,363	1	19,282
SA-29-LOI-23-P_1729379	a-23	68	1.21	3.64	81,690	1	-7,352
4102-2023-96_1758622	s-23	69	1.74	1.44	428,272	6	
44534730_1759916	s-23	70	1.13	1.40	138,180	1	-23,491
2023-09-B_1762324_Lot 01	s-23	71	1.29	1.18	1,015,338	2	
2023-09-B_1762324_Lot 02	s-23	71	1.28	1.31	1,355,536	1	-27,111
96900238_1768746	o-23	73	1.63	1.74	1,376,601	2	
QC-RAMQ-20230248_1768753	o-23	75	1.31	1.42	992,873	1	9,929
89065_1773518	o-23	76	1.93	1.26	199,082	7	
44534835_1773262	n-23	81	1.41	1.02	166,269	7	
44536368_1779110_Lot 03	n-23	82	1.68	1.33	198,585	6	
44536368_1779110_Lot 02	n-23	82	1.38	1.42	73,550	1	5,884
44536368_1779110_Lot 01	n-23	82	1.68	1.38	205,940	6	
95S10245_1781524_Lot 03	n-23	91	1.71	2.77	795,429	2	
95S10245_1781524_Lot 04	n-23	91	1.69	2.79	563,647	1	219,822
95S10245_1781524_Lot 01	n-23	91	1.71	2.87	578,477	1	237,176
95S10245_1781524_Lot 02	n-23	91	1.71	2.14	443,156	2	
AOP-0328_1787152	d-23	102	1.53	1.78	2,904,762	1	668,095
H23-1217MJR_1798198	j-24	104	1.38	1.52	485,951	11	
700001791_1807016	f-24	106	1.57	1.61	176,382	2	
700001792_1808261	f-24	107	1.48	1.24	1,380,991	3	
202403_1816375	m-24	110	1.75	1.51	621,410	7	
SECURITE-2024-2027_1831158_Lot 2	m-24	112	1.24	1.43	143,463	1	-8,608
SECURITE-2024-2027_1831158_Lot 1	m-24	112	1.24	1.43	173,894	1	-10,434
2024-55-64_1849263	m-24	117	1.62	1.10	1,396,347	2	
						Total	8,319,867.83
						Total with threshold	9,042,486.83

^a Dates are given as the initial of the month (as in the original French data) followed by the year.

We also observe an improvement in profits for the most recent contracts, which also corresponds to a decrease in the number of margins too low to be relevant. The model seems to better grasp how to approach the highest possible winning margin, without going for the lowest to win in all cases. We also note that for some contracts with several lots, the margins given are not all equal, which

clearly shows that the technical characteristics specific to the contracts are taken into account.

6.3.2 Comparison with the Commissionaires' reference behaviour

We now compare our model with the CCCQ's reference behaviour. To do so, we rely on 20 calls for tenders in which the CCCQ participated between 01/09/2023 and 01/04/2026. The oldest call for tenders studied considers 70 previous data points and the most recent considers 181. The results are reported in Table 6.3. The model's rank is calculated by considering the predicted margin as replacing the CCCQ's.

The "Predicted competitor precision" column shows the precision with which our tool identified competitors that were actually present in the call for tenders. In some cases, the lack of data does not allow this prediction. On average, the tool achieves 39% correct predictions.

If we consider the irrelevant contracts, i.e., the contracts for which the model predicts an unprofitable margin, only one case shows the model at a disadvantage compared with the CCCQ's choice (the CCCQ wins with a margin of 2.69 versus 1.08 for the model).

Overall, the model obtains a better rank than the CCCQ in 90% of cases; however, this must be nuanced by comparing the profits obtained. In terms of overall profit, the model outperforms the CCCQ, notably thanks to large gains on the lots of contract 2024-8023-50_20005431. On contract 2024-8023-50_20005431_LOT 06 - Volet C, won in both cases, the model improves the CCCQ's profits by about \$401,000. On the other contracts won by both the CCCQ and the model, the model incurs a profit loss of \$397,000, which cancels out the gain achieved previously.

If we consider only the calls for tenders where the margin given by the model seemed relevant, the model obtains a total profit of \$14,677,139 versus \$2,145,052 for the CCCQ. This is partly due to winning 6 additional contracts bringing in \$12,305,118. Without these, profits are equivalent, with \$2,372,020 for the model versus \$2,145,051 for the CCCQ. Note that a sensitivity analysis within a reasonable range (± 0.05) of the profitability threshold leads to the same results here.

6.4 Conclusion

The overall evaluation of our tool, carried out on both its prospecting and pricing prediction capabilities, confirms its relevance as a decision-support system for the CCCQ.

On the one hand, the prospecting component shows very satisfactory results. The Entity Matching algorithm proves robust, with a concordance rate of 92%, which automates and greatly strengthens the identification of contract history. In addition, the ability to anticipate renewals offers a strategic

advantage for preparing bids, even though it naturally remains subject to the vagaries of publication, notably contract options.

On the other hand, the analysis of the pricing model highlights crucial findings. While the raw algorithmic model sometimes tends to recommend extremely competitive margins, going as far as economic non-viability (below the ~ 0.3 threshold), the integration of a profitability threshold by the analysts effectively corrects this bias. Under these conditions, the model outperforms the CCCQ's historical decisions overall, obtaining a better ranking than the reference in most of the comparative scenarios studied and making it possible to win major contracts.

Although some features still have room for improvement (such as the exact prediction of competitors, at 39%), the model shows an excellent capacity to learn over time. Moreover, our solution should not be seen as a replacement for human expertise, but rather as a powerful optimization lever, capable of guiding the CCCQ towards maximizing its win rates and overall profitability.

Table 6.3 Model results for 20 calls for tenders compared with the CCCQ reference

Call for tenders	Model (margin)	CCCQ (margin)	Last winner (margin)	Predicted competitor precision	CCCQ rank	Model rank	Price	CCCQ profit	Model profit
44544014_20097223	1.16	1.46	1.21	42%	10	1	672,509		-94,151
AO-25-11-P-sp_20087874	1.08	2.69	2.69		1	1	138,092	191,593	-30,380
1000859_20087088	1.28	1.52	1.13	25%	8	4	3,964,947		
999741629-99741630_20081991_Lot 02	1.07	1.40	1.36	21%	3	1	23,097		-5,312
999741629-99741630_20081991_Lot 01	1.07	1.44	1.33	16%	2	1	49,317		-11,343
25-20987_20078343	1.57	1.80	1.53	20%	7	4	245,627		
2024/25-275_20052354	1.08	4.51	2.18	67%	6	1	172,488		-37,947
2025-8023-20_20041772_LOT 16 - Volet A	1.50	1.61	1.61	37%	3	2	12,039,234		
2025-8023-20_20041772_LOT 08 - Volet A	1.50	2.75	1.84	30%	9	1	3,093,698		618,740
2025-8023-20_20041772_LOT 05 - Volet A	1.50	1.62	1.52	33%	4	1	4,793,726		958,745
500955-2_20022357	2.55	2.73	2.60		6	2	2,054,924		
2024-569_20011888	1.41	1.72	1.72		1	1	65,404	27,754	7,194
2024-8023-50_20005431_LOT 06 - Volet C	1.60	1.54	1.61	38%	1	1	6,729,792	1,617,398	2,018,938
2024-8023-50_20005431_LOT 05 - Volet B	1.46	1.54	3.82	54%	2	1	24,359,194		3,897,471
2024-8023-50_20005431_LOT 04 - Volet B	1.46	1.54	3.81	54%	2	1	22,780,720		3,644,915
2024-8023-50_20005431_LOT 04 - Volet A	1.60	1.62	1.60	54%	2	1	10,617,491		3,185,247
2024-8023-50_20005431_LOT 03 - Volet C	1.62	1.54	1.34	46%	5	6	4,597,974		
44535338_1772515	1.41	1.46	1.46		1	1	3,144,440	499,899	345,888
AOP-0328_1787152	1.52	1.81	1.78		3	1	2,904,761		639,047
44534730_1759916	1.09	1.57	1.40	50%	6	1	138,179		-29,018
							Total	2,336,645	15,108,035
							Total with threshold		15,316,186

CHAPTER 7 CONCLUSION AND PERSPECTIVES

The literature review confirmed that, while modelling the bidding decision and pricing strategy is a field studied since the work of Friedman (1956) and Gates (1967), the joint integration of textual information extraction, similar-contract retrieval and pricing modelling remains poorly documented, particularly in the services sector and more specifically in security. Our scientific contribution lies precisely at this intersection. We proposed an integrated decision-support framework combining five complementary elements: an anticipation and historical mining module based on entity resolution; a textual data valorization module relying on a retrieval-augmented generation (RAG) architecture; a contract similarity module learned through ranking reinforcement; a Learning to Rank pricing model adapted to data scarcity; and an interactive visualization tool ensuring the coherence and usability of the whole. Applying these components together allows significant leveraging of historical data and provides systematic, quantitative indicators that support managers' decisions.

At the end of this work, it is worth returning to the objectives formulated in Chapter 3. The general objective was to improve the tender response process by leveraging historical data through the integration of artificial intelligence and data valorization methods. This objective was broken down into three specific objectives, which our work now allows us to consider achieved to different degrees. The first, characterizing the decision variables and indicators that structure bidding decisions, was fulfilled through the case study analysis and process modelling (Chapter 4), which highlighted the relevant indicators relating to both the contract and the competitive environment, and then through their systematic extraction using entity resolution and retrieval-augmented generation (RAG). The second, determining which models and approaches from the literature can be transposed to the partner's specific context, was addressed through the systematic review and critical analysis (Chapter 2), which led to selecting and then assembling a coherent set of approaches: entity resolution; RAG; learning to rank; and a Learning to Rank pricing model, all adapted to data scarcity and to the specifics of our case. The third, adapting the selected approaches and validating their viability in a real context by applying them to our partner's historical data, was satisfied through the development of the integrated tool (Chapter 5) and through the experiments carried out on five years of history (Chapter 6), which confirm the operational relevance of the whole with regard to the Commissionaires' internal reference.

Operationally, our work offers an integrated, usable tool, developed in Python and fed by both the SEAO open data and the partner's internal databases. The tool does not replace human expertise; it enriches it. It transforms the prospecting stance by offering a unified view of ongoing notices and anticipated renewals, automates the reading of technical specifications to systematically quantify a

contract’s material requirements, automatically retrieves contract history through entity resolution, and proposes a quantified margin recommendation based on similar contracts and recent market dynamics. The experiments carried out confirm the relevance of this approach: the Entity Matching algorithm achieves a very satisfactory concordance rate, the pricing model ranks among the top three bids in 70% of cases when restricted to economically viable margins, and it improves the CCCQ’s rank in the vast majority of the calls for tenders compared with the internal reference. Subject to an expert reading of the proposed margins, particularly when they fall below the economic viability threshold, the tool offers a substantial potential profit gain, while reducing the manual effort of prospecting and data searching.

Several limitations must nevertheless be acknowledged. First, the database used for training and evaluation remains modest. This scarcity constrains the choice of modelling architectures and limits the depth of the statistical analyses that can be carried out on performance. Furthermore, the indicators produced rely on historical data and therefore presuppose a certain stationarity of the market, i.e., that past dynamics remain valid for anticipating future decisions. While this assumption is a real limitation, it does not affect the value of the many descriptive analyses provided by the tool, which remain fully relevant in helping managers detect a possible change of contextual paradigm and reassess their decisions accordingly. Next, the model may suggest margins below the economic profitability threshold: without the manager’s mediation, the recommendations would be harmful. Also, the evaluation was conducted retrospectively on the Commissionaires’ history; it therefore does not fully predict the tool’s behaviour in continuous use, where its own recommendations would feed future decisions and, through a feedback effect, its own training base. Finally, the RAG extraction, although validated qualitatively and calibrated on the “uniforms” category, relies on a cost dictionary built with the help of experts, which would benefit from further study. In addition, this RAG infrastructure and method could be developed further in order to fully automate the extraction of technical data, thereby eliminating the last manual task needed to build the database.

In the short term, several avenues could strengthen the tool. Continuously enriching the internal database, by systematizing the archiving of working folders and retroactively completing contracts whose documents are incomplete, would increase the accuracy of both the similarity model and the pricing model. Integrating an explicit mechanism for managing the profitability threshold, or formulating Learning to Rank under a minimum profitability constraint, would also avoid non-viable recommendations. Competitor prediction could also be reformulated as a sequential modelling task that takes into account each player’s recent trajectory in the market, rather than as a static aggregate, but this would require more data on each competitor.

In the medium term, the tool would benefit from being deployed in production conditions, with appropriate change management support. Longitudinal monitoring of managers’ use of the tool, and

of their validations, adjustments and rejections of the recommendations, would provide valuable data for a feedback loop. An impact study, measuring changes in win rates and profitability before and after deployment, would also make it possible to objectively measure the operational benefit over a longer horizon.

In the longer term, two extensions seem particularly interesting. The first consists of extending the scope of the tool to other types of services (notably those offered by our partner, such as cybersecurity, investigation, surveillance, etc.) in order to test the generalization of the methodological framework beyond the security sector. The second concerns moving from a contract-by-contract response logic to a logic of managing a portfolio of calls for tenders, where the bidding choice and the margin would no longer be optimized independently, but by taking into account capacity constraints, strategic objectives and the aggregate risks of the portfolio. This opening would bring our work closer to recent project portfolio management approaches and would offer the CCCQ, or any similar organization, a coherent tool from upstream monitoring to strategic decision-making.

Ultimately, this thesis offers a concrete demonstration that it is possible, even with a modest volume of data and great documentary heterogeneity, to build a quantitative decision-support tool that is consistent with an expert reference on most of the indicators studied. The value of this contribution lies as much in the tool itself as in the approach: showing how natural language processing, learning-to-rank and visualization techniques can be articulated in a modular way to support a complex business process without sidelining human expertise. We hope that this articulation can inspire future work, both in the security services sector and in other tender response contexts.

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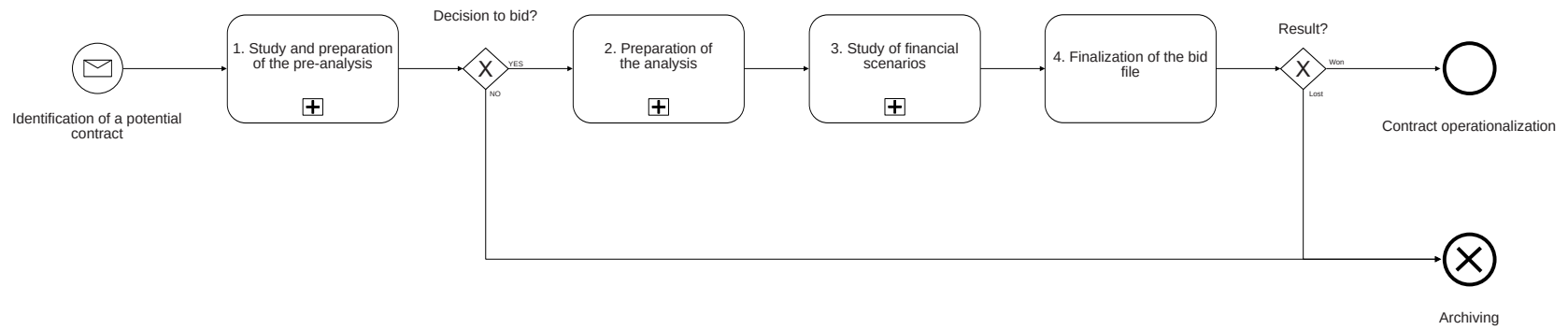
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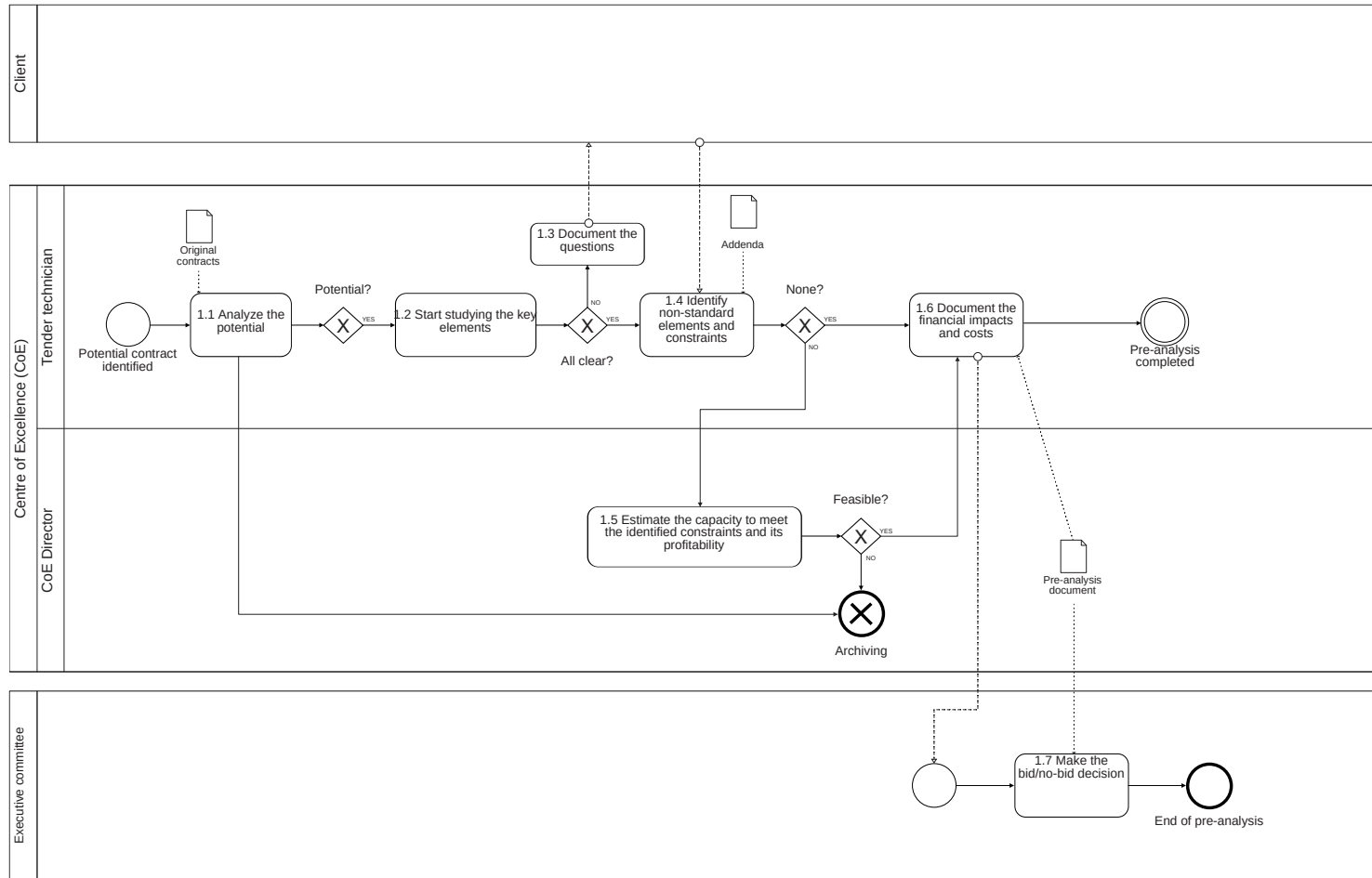
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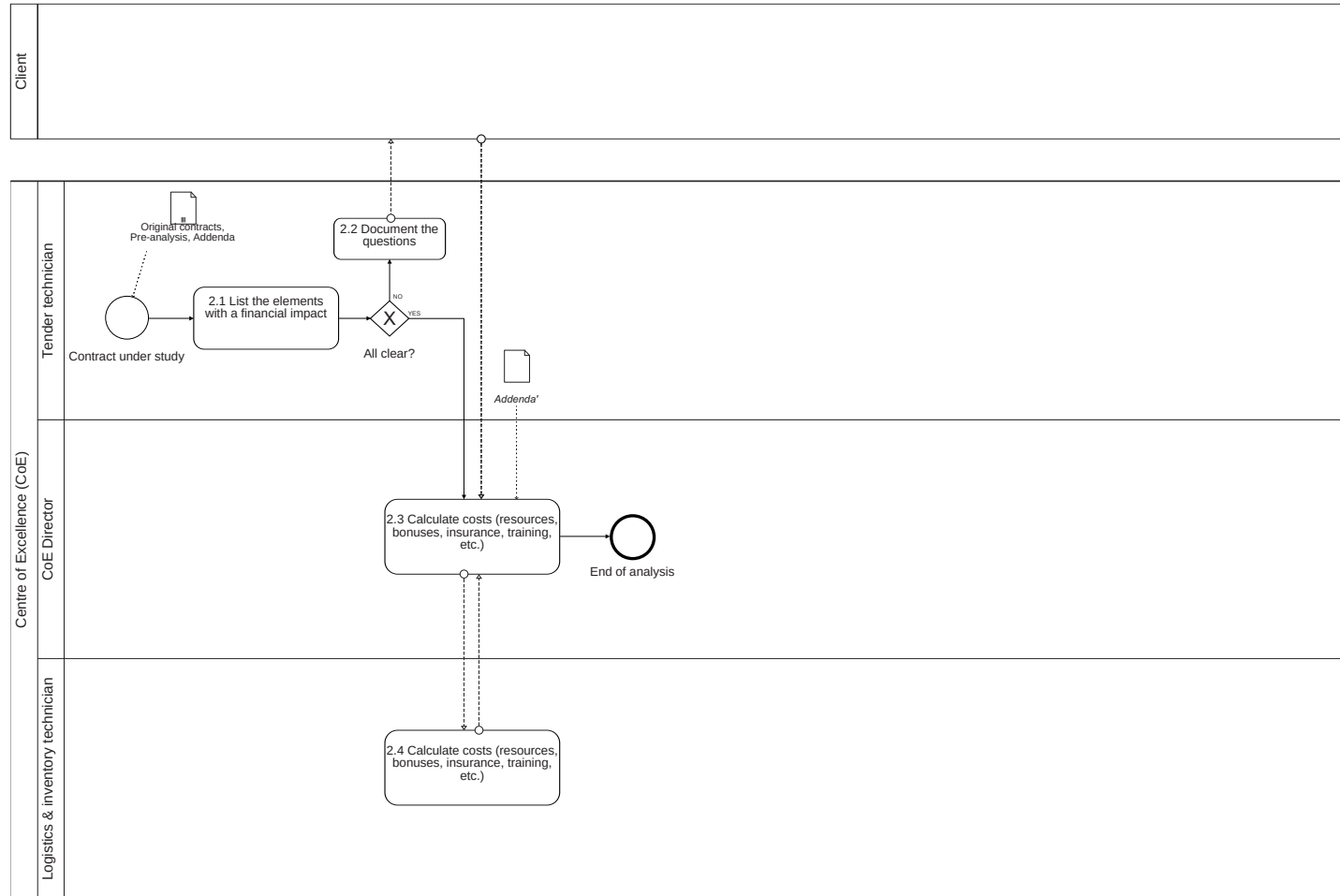
APPENDIX A GENERAL RESPONSE PROCESS (AS-IS)



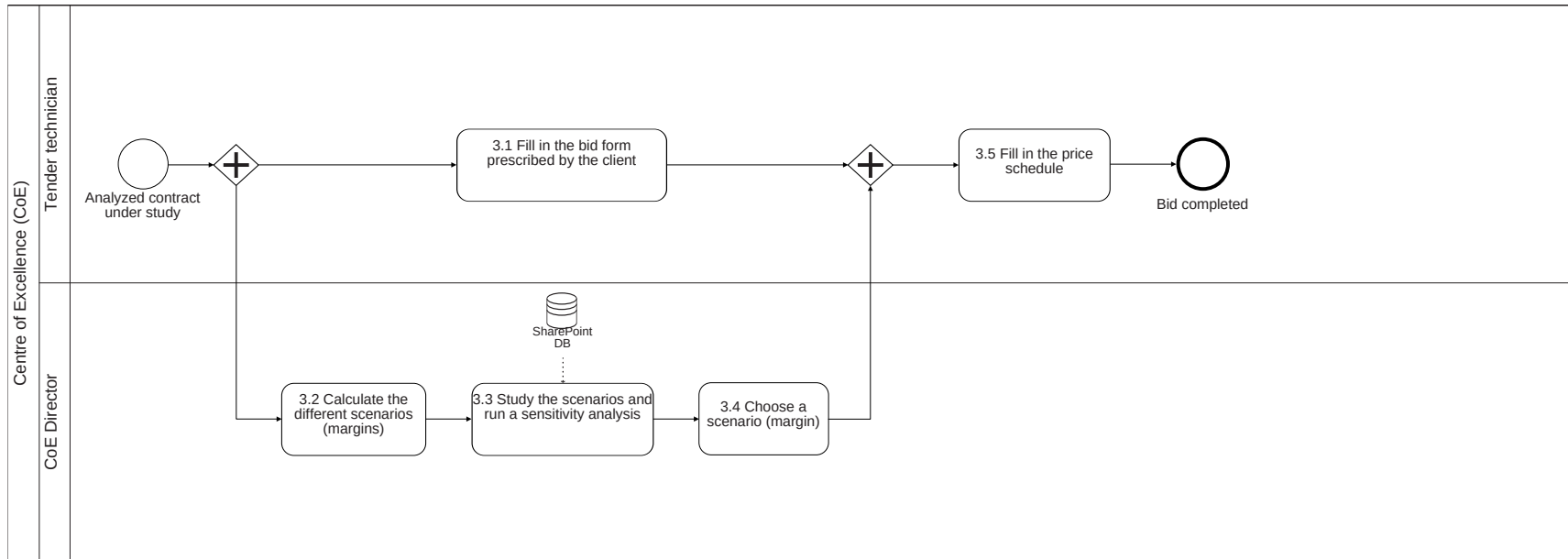
APPENDIX B RESPONSE PROCESS (AS-IS) PART 1: PRE-ANALYSIS



APPENDIX C RESPONSE PROCESS (AS-IS) PART 2: ANALYSIS

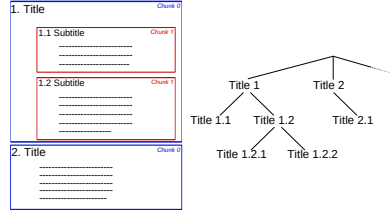


APPENDIX D RESPONSE PROCESS (AS-IS) PART 3: FINANCIAL SCENARIO



APPENDIX E SPARSE + HIERARCHY RETRIEVAL METHOD

1- CHUNKING



2- ENRICHED SPARSE EMBEDDINGS



Token_enriched = Token_base $\cup$ { D | $\exists t \in$ Token_base, $t \in V_D$ }, where V_D is the vocabulary of keyword D

3- RETRIEVAL

QUERY (Q)	Le prestataire doit-il fournir de l'équipement ?	['prestataire', 'doit', 'fournir', 'équipement']
CHUNK (C)	Le gardien doit apporter sa lampe mag-lite et sa radio.	['gardien', 'doit', 'apporter', 'lampe', 'mag-lite', 'radio', 'équipement', 'équipement']

3.1 Text Scoring Chunk

For a query Q and a chunk C :

$$Score(C, Q) = \sum_{i=1}^n IDF(q_i) \cdot \frac{f(q_i, C)(k_1 + 1)}{f(q_i, C) + k_1 \left(1 - b + b \frac{|C|}{avgdl}\right)}$$

$$IDF(q_i) = \ln \left(\frac{N - n(q_i) + 0.5}{n(q_i) + 0.5} + 1 \right)$$

$f(q_i, C)$ = number of occurrences of word q_i in chunk C
 $|C|$ = length of chunk C
 $avgdl$ = average length of the documents in the corpus
 N = number of documents in the corpus
 $n(q_i)$ = number of documents containing word q_i
 $b = 0.75$ (function parameter)
 $k_1 \in [1.2; 2.0]$ (optimization parameter)

→ **TF-IDF + Term saturation + Document length**

- **IDF**: word rarity factor (rare word = high weight)
- $f(q_i, C)(k_1 + 1)$: word frequency, with saturation introduced by k_1 (limits the weight of word repetition)
- $1 - b + b \frac{|C|}{avgdl}$: document length normalization (boosts the weight of shorter documents)

3.2 Hierarchical Scoring Chunk

For a query Q and a chunk C :

$$S_{base}(C) = Score(C, Q)$$

$$S_{parent}(C) = \begin{cases} S_{base}(P(C)) & \text{if } P(C) \neq \emptyset \\ S_{base}(C) & \text{otherwise} \end{cases}$$

$$S_{children}(C) = \begin{cases} \max_{e \in E(C)} S_{base}(e) & \text{if } E(C) \neq \emptyset \\ S_{base}(C) & \text{otherwise} \end{cases}$$

$$S(C) = S_{base}(C) \left(\frac{S_{parent}(C) + S_{children}(C)}{2} \right)$$

$K = \{ \text{the title of chunk } C \text{ contains a word from the dictionary } V_D \}$

3.3 Final Retrieval

$$Retrieved = \underbrace{\left\{ c \mid S(c) \geq \tau \cdot \max_{c \in C} S(c) \right\}}_{\text{Sparse retrieval}} \cup \underbrace{\left\{ k \mid S(k) \geq \beta \cdot \max_{c \in C} S(c) \text{ if } k \in K \right\}}_{\text{Naive retrieval}}$$

with τ and β defined thresholds